



STEP-1: EXPLORATORY DATA ANALYSIS

In [2]: *# STEP 1: IMPORTING REQUIRED LIBRARIES*

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [3]: *# STEP 2: SET DISPLAY OPTION*

```
pd.set_option("display.max_columns", None)
pd.set_option('display.max_rows', None)
```

In [4]: *# STEP 3: LOADING THE DATASET*

```
df=pd.read_csv("concrete_data.csv")
```

In [5]: df

Out[5]:

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_agg
0	540.0	0.0	0.0	162.0	2.5	1
1	540.0	0.0	0.0	162.0	2.5	1
2	332.5	142.5	0.0	228.0	0.0	
3	332.5	142.5	0.0	228.0	0.0	
4	198.6	132.4	0.0	192.0	0.0	
5	266.0	114.0	0.0	228.0	0.0	
6	380.0	95.0	0.0	228.0	0.0	
7	380.0	95.0	0.0	228.0	0.0	
8	266.0	114.0	0.0	228.0	0.0	
9	475.0	0.0	0.0	228.0	0.0	
10	198.6	132.4	0.0	192.0	0.0	
11	198.6	132.4	0.0	192.0	0.0	
12	427.5	47.5	0.0	228.0	0.0	
13	190.0	190.0	0.0	228.0	0.0	
14	304.0	76.0	0.0	228.0	0.0	
15	380.0	0.0	0.0	228.0	0.0	
16	139.6	209.4	0.0	192.0	0.0	1
17	342.0	38.0	0.0	228.0	0.0	
18	380.0	95.0	0.0	228.0	0.0	
19	475.0	0.0	0.0	228.0	0.0	
20	427.5	47.5	0.0	228.0	0.0	
21	139.6	209.4	0.0	192.0	0.0	1
22	139.6	209.4	0.0	192.0	0.0	1
23	139.6	209.4	0.0	192.0	0.0	1
24	380.0	0.0	0.0	228.0	0.0	
25	380.0	0.0	0.0	228.0	0.0	
26	380.0	95.0	0.0	228.0	0.0	
27	342.0	38.0	0.0	228.0	0.0	
28	427.5	47.5	0.0	228.0	0.0	
29	475.0	0.0	0.0	228.0	0.0	
30	304.0	76.0	0.0	228.0	0.0	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_agg
31	266.0	114.0	0.0	228.0	0.0	
32	198.6	132.4	0.0	192.0	0.0	
33	475.0	0.0	0.0	228.0	0.0	
34	190.0	190.0	0.0	228.0	0.0	
35	237.5	237.5	0.0	228.0	0.0	
36	237.5	237.5	0.0	228.0	0.0	
37	332.5	142.5	0.0	228.0	0.0	
38	475.0	0.0	0.0	228.0	0.0	
39	237.5	237.5	0.0	228.0	0.0	
40	342.0	38.0	0.0	228.0	0.0	
41	427.5	47.5	0.0	228.0	0.0	
42	237.5	237.5	0.0	228.0	0.0	
43	380.0	0.0	0.0	228.0	0.0	
44	427.5	47.5	0.0	228.0	0.0	
45	427.5	47.5	0.0	228.0	0.0	
46	349.0	0.0	0.0	192.0	0.0	1
47	380.0	95.0	0.0	228.0	0.0	
48	237.5	237.5	0.0	228.0	0.0	
49	380.0	95.0	0.0	228.0	0.0	
50	332.5	142.5	0.0	228.0	0.0	
51	190.0	190.0	0.0	228.0	0.0	
52	237.5	237.5	0.0	228.0	0.0	
53	304.0	76.0	0.0	228.0	0.0	
54	139.6	209.4	0.0	192.0	0.0	1
55	198.6	132.4	0.0	192.0	0.0	
56	475.0	0.0	0.0	228.0	0.0	
57	198.6	132.4	0.0	192.0	0.0	
58	304.0	76.0	0.0	228.0	0.0	
59	332.5	142.5	0.0	228.0	0.0	
60	304.0	76.0	0.0	228.0	0.0	
61	266.0	114.0	0.0	228.0	0.0	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
62	310.0	0.0	0.0	192.0	0.0	
63	190.0	190.0	0.0	228.0	0.0	
64	266.0	114.0	0.0	228.0	0.0	
65	342.0	38.0	0.0	228.0	0.0	
66	139.6	209.4	0.0	192.0	0.0	1
67	332.5	142.5	0.0	228.0	0.0	
68	190.0	190.0	0.0	228.0	0.0	
69	485.0	0.0	0.0	146.0	0.0	1
70	374.0	189.2	0.0	170.1	10.1	
71	313.3	262.2	0.0	175.5	8.6	1
72	425.0	106.3	0.0	153.5	16.5	
73	425.0	106.3	0.0	151.4	18.6	
74	375.0	93.8	0.0	126.6	23.4	
75	475.0	118.8	0.0	181.1	8.9	
76	469.0	117.2	0.0	137.8	32.2	
77	425.0	106.3	0.0	153.5	16.5	
78	388.6	97.1	0.0	157.9	12.1	
79	531.3	0.0	0.0	141.8	28.2	
80	425.0	106.3	0.0	153.5	16.5	
81	318.8	212.5	0.0	155.7	14.3	
82	401.8	94.7	0.0	147.4	11.4	
83	362.6	189.0	0.0	164.9	11.6	
84	323.7	282.8	0.0	183.8	10.3	
85	379.5	151.2	0.0	153.9	15.9	1
86	362.6	189.0	0.0	164.9	11.6	
87	286.3	200.9	0.0	144.7	11.2	1
88	362.6	189.0	0.0	164.9	11.6	
89	439.0	177.0	0.0	186.0	11.1	
90	389.9	189.0	0.0	145.9	22.0	
91	362.6	189.0	0.0	164.9	11.6	
92	337.9	189.0	0.0	174.9	9.5	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_agg
93	374.0	189.2	0.0	170.1	10.1	
94	313.3	262.2	0.0	175.5	8.6	1
95	425.0	106.3	0.0	153.5	16.5	
96	425.0	106.3	0.0	151.4	18.6	
97	375.0	93.8	0.0	126.6	23.4	
98	475.0	118.8	0.0	181.1	8.9	
99	469.0	117.2	0.0	137.8	32.2	
100	425.0	106.3	0.0	153.5	16.5	
101	388.6	97.1	0.0	157.9	12.1	
102	531.3	0.0	0.0	141.8	28.2	
103	425.0	106.3	0.0	153.5	16.5	
104	318.8	212.5	0.0	155.7	14.3	
105	401.8	94.7	0.0	147.4	11.4	
106	362.6	189.0	0.0	164.9	11.6	
107	323.7	282.8	0.0	183.8	10.3	
108	379.5	151.2	0.0	153.9	15.9	1
109	362.6	189.0	0.0	164.9	11.6	
110	286.3	200.9	0.0	144.7	11.2	1
111	362.6	189.0	0.0	164.9	11.6	
112	439.0	177.0	0.0	186.0	11.1	
113	389.9	189.0	0.0	145.9	22.0	
114	362.6	189.0	0.0	164.9	11.6	
115	337.9	189.0	0.0	174.9	9.5	
116	374.0	189.2	0.0	170.1	10.1	
117	313.3	262.2	0.0	175.5	8.6	1
118	425.0	106.3	0.0	153.5	16.5	
119	425.0	106.3	0.0	151.4	18.6	
120	375.0	93.8	0.0	126.6	23.4	
121	475.0	118.8	0.0	181.1	8.9	
122	469.0	117.2	0.0	137.8	32.2	
123	425.0	106.3	0.0	153.5	16.5	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
124	388.6	97.1	0.0	157.9	12.1	
125	531.3	0.0	0.0	141.8	28.2	
126	425.0	106.3	0.0	153.5	16.5	
127	318.8	212.5	0.0	155.7	14.3	
128	401.8	94.7	0.0	147.4	11.4	
129	362.6	189.0	0.0	164.9	11.6	
130	323.7	282.8	0.0	183.8	10.3	
131	379.5	151.2	0.0	153.9	15.9	1
132	362.6	189.0	0.0	164.9	11.6	
133	286.3	200.9	0.0	144.7	11.2	1
134	362.6	189.0	0.0	164.9	11.6	
135	439.0	177.0	0.0	186.0	11.1	
136	389.9	189.0	0.0	145.9	22.0	
137	362.6	189.0	0.0	164.9	11.6	
138	337.9	189.0	0.0	174.9	9.5	
139	374.0	189.2	0.0	170.1	10.1	
140	313.3	262.2	0.0	175.5	8.6	1
141	425.0	106.3	0.0	153.5	16.5	
142	425.0	106.3	0.0	151.4	18.6	
143	375.0	93.8	0.0	126.6	23.4	
144	475.0	118.8	0.0	181.1	8.9	
145	469.0	117.2	0.0	137.8	32.2	
146	425.0	106.3	0.0	153.5	16.5	
147	388.6	97.1	0.0	157.9	12.1	
148	531.3	0.0	0.0	141.8	28.2	
149	425.0	106.3	0.0	153.5	16.5	
150	318.8	212.5	0.0	155.7	14.3	
151	401.8	94.7	0.0	147.4	11.4	
152	362.6	189.0	0.0	164.9	11.6	
153	323.7	282.8	0.0	183.8	10.3	
154	379.5	151.2	0.0	153.9	15.9	1

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
155	362.6	189.0	0.0	164.9	11.6	
156	286.3	200.9	0.0	144.7	11.2	1
157	362.6	189.0	0.0	164.9	11.6	
158	439.0	177.0	0.0	186.0	11.1	
159	389.9	189.0	0.0	145.9	22.0	
160	362.6	189.0	0.0	164.9	11.6	
161	337.9	189.0	0.0	174.9	9.5	
162	374.0	189.2	0.0	170.1	10.1	
163	313.3	262.2	0.0	175.5	8.6	1
164	425.0	106.3	0.0	153.5	16.5	
165	425.0	106.3	0.0	151.4	18.6	
166	375.0	93.8	0.0	126.6	23.4	
167	475.0	118.8	0.0	181.1	8.9	
168	469.0	117.2	0.0	137.8	32.2	
169	425.0	106.3	0.0	153.5	16.5	
170	388.6	97.1	0.0	157.9	12.1	
171	531.3	0.0	0.0	141.8	28.2	
172	425.0	106.3	0.0	153.5	16.5	
173	318.8	212.5	0.0	155.7	14.3	
174	401.8	94.7	0.0	147.4	11.4	
175	362.6	189.0	0.0	164.9	11.6	
176	379.5	151.2	0.0	153.9	15.9	1
177	362.6	189.0	0.0	164.9	11.6	
178	286.3	200.9	0.0	144.7	11.2	1
179	362.6	189.0	0.0	164.9	11.6	
180	439.0	177.0	0.0	186.0	11.1	
181	389.9	189.0	0.0	145.9	22.0	
182	362.6	189.0	0.0	164.9	11.6	
183	337.9	189.0	0.0	174.9	9.5	
184	222.4	0.0	96.7	189.3	4.5	
185	222.4	0.0	96.7	189.3	4.5	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
186	222.4	0.0	96.7	189.3	4.5	
187	222.4	0.0	96.7	189.3	4.5	
188	222.4	0.0	96.7	189.3	4.5	
189	233.8	0.0	94.6	197.9	4.6	
190	233.8	0.0	94.6	197.9	4.6	
191	233.8	0.0	94.6	197.9	4.6	
192	233.8	0.0	94.6	197.9	4.6	
193	233.8	0.0	94.6	197.9	4.6	
194	194.7	0.0	100.5	165.6	7.5	1
195	194.7	0.0	100.5	165.6	7.5	1
196	194.7	0.0	100.5	165.6	7.5	1
197	194.7	0.0	100.5	165.6	7.5	1
198	194.7	0.0	100.5	165.6	7.5	1
199	190.7	0.0	125.4	162.1	7.8	1
200	190.7	0.0	125.4	162.1	7.8	1
201	190.7	0.0	125.4	162.1	7.8	1
202	190.7	0.0	125.4	162.1	7.8	1
203	190.7	0.0	125.4	162.1	7.8	1
204	212.1	0.0	121.6	180.3	5.7	1
205	212.1	0.0	121.6	180.3	5.7	1
206	212.1	0.0	121.6	180.3	5.7	1
207	212.1	0.0	121.6	180.3	5.7	1
208	212.1	0.0	121.6	180.3	5.7	1
209	230.0	0.0	118.3	195.5	4.6	1
210	230.0	0.0	118.3	195.5	4.6	1
211	230.0	0.0	118.3	195.5	4.6	1
212	230.0	0.0	118.3	195.5	4.6	1
213	230.0	0.0	118.3	195.5	4.6	1
214	190.3	0.0	125.2	161.9	9.9	1
215	190.3	0.0	125.2	161.9	9.9	1
216	190.3	0.0	125.2	161.9	9.9	1

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
217	190.3	0.0	125.2	161.9	9.9	1
218	190.3	0.0	125.2	161.9	9.9	1
219	166.1	0.0	163.3	176.5	4.5	1
220	166.1	0.0	163.3	176.5	4.5	1
221	166.1	0.0	163.3	176.5	4.5	1
222	166.1	0.0	163.3	176.5	4.5	1
223	166.1	0.0	163.3	176.5	4.5	1
224	168.0	42.1	163.8	121.8	5.7	1
225	168.0	42.1	163.8	121.8	5.7	1
226	168.0	42.1	163.8	121.8	5.7	1
227	168.0	42.1	163.8	121.8	5.7	1
228	168.0	42.1	163.8	121.8	5.7	1
229	213.7	98.1	24.5	181.7	6.9	1
230	213.7	98.1	24.5	181.7	6.9	1
231	213.7	98.1	24.5	181.7	6.9	1
232	213.7	98.1	24.5	181.7	6.9	1
233	213.7	98.1	24.5	181.7	6.9	1
234	213.8	98.1	24.5	181.7	6.7	1
235	213.8	98.1	24.5	181.7	6.7	1
236	213.8	98.1	24.5	181.7	6.7	1
237	213.8	98.1	24.5	181.7	6.7	1
238	213.8	98.1	24.5	181.7	6.7	1
239	229.7	0.0	118.2	195.2	6.1	1
240	229.7	0.0	118.2	195.2	6.1	1
241	229.7	0.0	118.2	195.2	6.1	1
242	229.7	0.0	118.2	195.2	6.1	1
243	229.7	0.0	118.2	195.2	6.1	1
244	238.1	0.0	94.1	186.7	7.0	
245	238.1	0.0	94.1	186.7	7.0	
246	238.1	0.0	94.1	186.7	7.0	
247	238.1	0.0	94.1	186.7	7.0	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
248	238.1	0.0	94.1	186.7	7.0	
249	250.0	0.0	95.7	187.4	5.5	
250	250.0	0.0	95.7	187.4	5.5	
251	250.0	0.0	95.7	187.4	5.5	
252	250.0	0.0	95.7	187.4	5.5	
253	250.0	0.0	95.7	187.4	5.5	
254	212.5	0.0	100.4	159.3	8.7	1
255	212.5	0.0	100.4	159.3	8.7	1
256	212.5	0.0	100.4	159.3	8.7	1
257	212.5	0.0	100.4	159.3	8.7	1
258	212.5	0.0	100.4	159.3	8.7	1
259	212.6	0.0	100.4	159.4	10.4	1
260	212.6	0.0	100.4	159.4	10.4	1
261	212.6	0.0	100.4	159.4	10.4	1
262	212.6	0.0	100.4	159.4	10.4	1
263	212.6	0.0	100.4	159.4	10.4	1
264	212.0	0.0	124.8	159.0	7.8	1
265	212.0	0.0	124.8	159.0	7.8	1
266	212.0	0.0	124.8	159.0	7.8	1
267	212.0	0.0	124.8	159.0	7.8	1
268	212.0	0.0	124.8	159.0	7.8	1
269	231.8	0.0	121.6	174.0	6.7	1
270	231.8	0.0	121.6	174.0	6.7	1
271	231.8	0.0	121.6	174.0	6.7	1
272	231.8	0.0	121.6	174.0	6.7	1
273	231.8	0.0	121.6	174.0	6.7	1
274	251.4	0.0	118.3	188.5	5.8	1
275	251.4	0.0	118.3	188.5	5.8	1
276	251.4	0.0	118.3	188.5	5.8	1
277	251.4	0.0	118.3	188.5	5.8	1
278	251.4	0.0	118.3	188.5	5.8	1

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
279	251.4	0.0	118.3	188.5	6.4	1
280	251.4	0.0	118.3	188.5	6.4	1
281	251.4	0.0	118.3	188.5	6.4	1
282	251.4	0.0	118.3	188.5	6.4	1
283	251.4	0.0	118.3	188.5	6.4	1
284	181.4	0.0	167.0	169.6	7.6	1
285	181.4	0.0	167.0	169.6	7.6	1
286	181.4	0.0	167.0	169.6	7.6	1
287	181.4	0.0	167.0	169.6	7.6	1
288	181.4	0.0	167.0	169.6	7.6	1
289	182.0	45.2	122.0	170.2	8.2	1
290	182.0	45.2	122.0	170.2	8.2	1
291	182.0	45.2	122.0	170.2	8.2	1
292	182.0	45.2	122.0	170.2	8.2	1
293	182.0	45.2	122.0	170.2	8.2	1
294	168.9	42.2	124.3	158.3	10.8	1
295	168.9	42.2	124.3	158.3	10.8	1
296	168.9	42.2	124.3	158.3	10.8	1
297	168.9	42.2	124.3	158.3	10.8	1
298	168.9	42.2	124.3	158.3	10.8	1
299	290.4	0.0	96.2	168.1	9.4	
300	290.4	0.0	96.2	168.1	9.4	
301	290.4	0.0	96.2	168.1	9.4	
302	290.4	0.0	96.2	168.1	9.4	
303	290.4	0.0	96.2	168.1	9.4	
304	277.1	0.0	97.4	160.6	11.8	
305	277.1	0.0	97.4	160.6	11.8	
306	277.1	0.0	97.4	160.6	11.8	
307	277.1	0.0	97.4	160.6	11.8	
308	277.1	0.0	97.4	160.6	11.8	
309	295.7	0.0	95.6	171.5	8.9	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
310	295.7	0.0	95.6	171.5	8.9	
311	295.7	0.0	95.6	171.5	8.9	
312	295.7	0.0	95.6	171.5	8.9	
313	295.7	0.0	95.6	171.5	8.9	
314	251.8	0.0	99.9	146.1	12.4	1
315	251.8	0.0	99.9	146.1	12.4	1
316	251.8	0.0	99.9	146.1	12.4	1
317	251.8	0.0	99.9	146.1	12.4	1
318	251.8	0.0	99.9	146.1	12.4	1
319	249.1	0.0	98.8	158.1	12.8	
320	249.1	0.0	98.8	158.1	12.8	
321	249.1	0.0	98.8	158.1	12.8	
322	249.1	0.0	98.8	158.1	12.8	
323	249.1	0.0	98.8	158.1	12.8	
324	252.3	0.0	98.8	146.3	14.2	
325	252.3	0.0	98.8	146.3	14.2	
326	252.3	0.0	98.8	146.3	14.2	
327	252.3	0.0	98.8	146.3	14.2	
328	252.3	0.0	98.8	146.3	14.2	
329	246.8	0.0	125.1	143.3	12.0	1
330	246.8	0.0	125.1	143.3	12.0	1
331	246.8	0.0	125.1	143.3	12.0	1
332	246.8	0.0	125.1	143.3	12.0	1
333	246.8	0.0	125.1	143.3	12.0	1
334	275.1	0.0	121.4	159.5	9.9	1
335	275.1	0.0	121.4	159.5	9.9	1
336	275.1	0.0	121.4	159.5	9.9	1
337	275.1	0.0	121.4	159.5	9.9	1
338	275.1	0.0	121.4	159.5	9.9	1
339	297.2	0.0	117.5	174.8	9.5	1
340	297.2	0.0	117.5	174.8	9.5	1

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
341	297.2	0.0	117.5	174.8	9.5	1
342	297.2	0.0	117.5	174.8	9.5	1
343	297.2	0.0	117.5	174.8	9.5	1
344	213.7	0.0	174.7	154.8	10.2	1
345	213.7	0.0	174.7	154.8	10.2	1
346	213.7	0.0	174.7	154.8	10.2	1
347	213.7	0.0	174.7	154.8	10.2	1
348	213.7	0.0	174.7	154.8	10.2	1
349	213.5	0.0	174.2	154.6	11.7	1
350	213.5	0.0	174.2	154.6	11.7	1
351	213.5	0.0	174.2	154.6	11.7	1
352	213.5	0.0	174.2	154.6	11.7	1
353	213.5	0.0	174.2	154.6	11.7	1
354	277.2	97.8	24.5	160.7	11.2	1
355	277.2	97.8	24.5	160.7	11.2	1
356	277.2	97.8	24.5	160.7	11.2	1
357	277.2	97.8	24.5	160.7	11.2	1
358	277.2	97.8	24.5	160.7	11.2	1
359	218.2	54.6	123.8	140.8	11.9	1
360	218.2	54.6	123.8	140.8	11.9	1
361	218.2	54.6	123.8	140.8	11.9	1
362	218.2	54.6	123.8	140.8	11.9	1
363	218.2	54.6	123.8	140.8	11.9	1
364	214.9	53.8	121.9	155.6	9.6	1
365	214.9	53.8	121.9	155.6	9.6	1
366	214.9	53.8	121.9	155.6	9.6	1
367	214.9	53.8	121.9	155.6	9.6	1
368	214.9	53.8	121.9	155.6	9.6	1
369	218.9	0.0	124.1	158.5	11.3	1
370	218.9	0.0	124.1	158.5	11.3	1
371	218.9	0.0	124.1	158.5	11.3	1

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
372	218.9	0.0	124.1	158.5	11.3	1
373	218.9	0.0	124.1	158.5	11.3	1
374	376.0	0.0	0.0	214.6	0.0	1
375	376.0	0.0	0.0	214.6	0.0	1
376	376.0	0.0	0.0	214.6	0.0	1
377	376.0	0.0	0.0	214.6	0.0	1
378	376.0	0.0	0.0	214.6	0.0	1
379	500.0	0.0	0.0	140.0	4.0	
380	475.0	0.0	59.0	142.0	1.9	1
381	315.0	137.0	0.0	145.0	5.9	1
382	505.0	0.0	60.0	195.0	0.0	1
383	451.0	0.0	0.0	165.0	11.3	1
384	516.0	0.0	0.0	162.0	8.2	
385	520.0	0.0	0.0	170.0	5.2	
386	528.0	0.0	0.0	185.0	6.9	
387	520.0	0.0	0.0	175.0	5.2	
388	385.0	0.0	136.0	158.0	20.0	
389	500.1	0.0	0.0	200.0	3.0	1
390	450.1	50.0	0.0	200.0	3.0	1
391	397.0	17.2	158.0	167.0	20.8	
392	333.0	17.5	163.0	167.0	17.9	
393	334.0	17.6	158.0	189.0	15.3	
394	405.0	0.0	0.0	175.0	0.0	1
395	200.0	200.0	0.0	190.0	0.0	1
396	516.0	0.0	0.0	162.0	8.3	
397	145.0	116.0	119.0	184.0	5.7	
398	160.0	128.0	122.0	182.0	6.4	
399	234.0	156.0	0.0	189.0	5.9	
400	250.0	180.0	95.0	159.0	9.5	
401	475.0	0.0	0.0	162.0	9.5	1
402	285.0	190.0	0.0	163.0	7.6	1

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_agg
403	356.0	119.0	0.0	160.0	9.0	1
404	275.0	180.0	120.0	162.0	10.4	
405	500.0	0.0	0.0	151.0	9.0	1
406	165.0	0.0	143.6	163.8	0.0	1
407	165.0	128.5	132.1	175.1	8.1	1
408	178.0	129.8	118.6	179.9	3.6	1
409	167.4	129.9	128.6	175.5	7.8	1
410	172.4	13.6	172.4	156.8	4.1	1
411	173.5	50.1	173.5	164.8	6.5	1
412	167.0	75.4	167.0	164.0	7.9	1
413	173.8	93.4	159.9	172.3	9.7	1
414	190.3	0.0	125.2	166.6	9.9	1
415	250.0	0.0	95.7	191.8	5.3	
416	213.5	0.0	174.2	159.2	11.7	1
417	194.7	0.0	100.5	170.2	7.5	
418	251.4	0.0	118.3	192.9	5.8	1
419	165.0	0.0	143.6	163.8	0.0	1
420	165.0	128.5	132.1	175.1	8.1	1
421	178.0	129.8	118.6	179.9	3.6	1
422	167.4	129.9	128.6	175.5	7.8	1
423	172.4	13.6	172.4	156.8	4.1	1
424	173.5	50.1	173.5	164.8	6.5	1
425	167.0	75.4	167.0	164.0	7.9	1
426	173.8	93.4	159.9	172.3	9.7	1
427	190.3	0.0	125.2	166.6	9.9	1
428	250.0	0.0	95.7	191.8	5.3	
429	213.5	0.0	174.2	159.2	11.7	1
430	194.7	0.0	100.5	170.2	7.5	
431	251.4	0.0	118.3	192.9	5.8	1
432	165.0	0.0	143.6	163.8	0.0	1
433	165.0	128.5	132.1	175.1	8.1	1

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
434	178.0	129.8	118.6	179.9	3.6	1
435	167.4	129.9	128.6	175.5	7.8	1
436	172.4	13.6	172.4	156.8	4.1	1
437	173.5	50.1	173.5	164.8	6.5	1
438	167.0	75.4	167.0	164.0	7.9	1
439	173.8	93.4	159.9	172.3	9.7	1
440	190.3	0.0	125.2	166.6	9.9	1
441	250.0	0.0	95.7	191.8	5.3	
442	213.5	0.0	174.2	159.2	11.7	1
443	194.7	0.0	100.5	170.2	7.5	
444	251.4	0.0	118.3	192.9	5.8	1
445	165.0	0.0	143.6	163.8	0.0	1
446	165.0	128.5	132.1	175.1	8.1	1
447	178.0	129.8	118.6	179.9	3.6	1
448	167.4	129.9	128.6	175.5	7.8	1
449	172.4	13.6	172.4	156.8	4.1	1
450	173.5	50.1	173.5	164.8	6.5	1
451	167.0	75.4	167.0	164.0	7.9	1
452	173.8	93.4	159.9	172.3	9.7	1
453	190.3	0.0	125.2	166.6	9.9	1
454	250.0	0.0	95.7	191.8	5.3	
455	213.5	0.0	174.2	159.2	11.7	1
456	194.7	0.0	100.5	170.2	7.5	
457	251.4	0.0	118.3	192.9	5.8	1
458	165.0	0.0	143.6	163.8	0.0	1
459	165.0	128.5	132.1	175.1	8.1	1
460	178.0	129.8	118.6	179.9	3.6	1
461	167.4	129.9	128.6	175.5	7.8	1
462	172.4	13.6	172.4	156.8	4.1	1
463	173.5	50.1	173.5	164.8	6.5	1
464	167.0	75.4	167.0	164.0	7.9	1

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
465	173.8	93.4	159.9	172.3	9.7	1
466	190.3	0.0	125.2	166.6	9.9	1
467	250.0	0.0	95.7	191.8	5.3	
468	213.5	0.0	174.2	159.2	11.7	1
469	194.7	0.0	100.5	170.2	7.5	
470	251.4	0.0	118.3	192.9	5.8	1
471	446.0	24.0	79.0	162.0	11.6	
472	446.0	24.0	79.0	162.0	11.6	
473	446.0	24.0	79.0	162.0	11.6	
474	446.0	24.0	79.0	162.0	10.3	
475	446.0	24.0	79.0	162.0	11.6	
476	446.0	24.0	79.0	162.0	11.6	
477	446.0	24.0	79.0	162.0	11.6	
478	446.0	24.0	79.0	162.0	11.6	
479	446.0	24.0	79.0	162.0	11.6	
480	446.0	24.0	79.0	162.0	11.6	
481	446.0	24.0	79.0	162.0	11.6	
482	446.0	24.0	79.0	162.0	11.6	
483	446.0	24.0	79.0	162.0	11.6	
484	446.0	24.0	79.0	162.0	10.3	
485	387.0	20.0	94.0	157.0	14.3	
486	387.0	20.0	94.0	157.0	13.9	
487	387.0	20.0	94.0	157.0	11.6	
488	387.0	20.0	94.0	157.0	14.3	
489	387.0	20.0	94.0	157.0	13.9	
490	387.0	20.0	94.0	157.0	11.6	
491	387.0	20.0	94.0	157.0	14.3	
492	387.0	20.0	94.0	157.0	13.9	
493	387.0	20.0	94.0	157.0	11.6	
494	387.0	20.0	94.0	157.0	14.3	
495	387.0	20.0	94.0	157.0	13.9	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
496	387.0	20.0	94.0	157.0	11.6	
497	355.0	19.0	97.0	145.0	13.1	
498	355.0	19.0	97.0	145.0	12.3	
499	491.0	26.0	123.0	210.0	3.9	
500	491.0	26.0	123.0	201.0	3.9	
501	491.0	26.0	123.0	210.0	3.9	
502	491.0	26.0	123.0	210.0	3.9	
503	491.0	26.0	123.0	210.0	3.9	
504	491.0	26.0	123.0	201.0	3.9	
505	491.0	26.0	123.0	201.0	3.9	
506	491.0	26.0	123.0	201.0	3.9	
507	424.0	22.0	132.0	178.0	8.5	
508	424.0	22.0	132.0	178.0	8.5	
509	424.0	22.0	132.0	168.0	8.9	
510	424.0	22.0	132.0	178.0	8.5	
511	424.0	22.0	132.0	178.0	8.5	
512	424.0	22.0	132.0	168.0	8.9	
513	424.0	22.0	132.0	168.0	8.9	
514	424.0	22.0	132.0	168.0	8.9	
515	202.0	11.0	141.0	206.0	1.7	
516	202.0	11.0	141.0	206.0	1.7	
517	202.0	11.0	141.0	206.0	1.7	
518	202.0	11.0	141.0	206.0	1.7	
519	284.0	15.0	141.0	179.0	5.5	
520	284.0	15.0	141.0	179.0	5.5	
521	284.0	15.0	141.0	179.0	5.5	
522	284.0	15.0	141.0	179.0	5.5	
523	359.0	19.0	141.0	154.0	10.9	
524	359.0	19.0	141.0	154.0	10.9	
525	359.0	19.0	141.0	154.0	10.9	
526	359.0	19.0	141.0	154.0	10.9	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
527	359.0	19.0	141.0	154.0	10.9	
528	359.0	19.0	141.0	154.0	10.9	
529	359.0	19.0	141.0	154.0	10.9	
530	359.0	19.0	141.0	154.0	10.9	
531	436.0	0.0	0.0	218.0	0.0	
532	289.0	0.0	0.0	192.0	0.0	
533	289.0	0.0	0.0	192.0	0.0	
534	393.0	0.0	0.0	192.0	0.0	
535	393.0	0.0	0.0	192.0	0.0	
536	393.0	0.0	0.0	192.0	0.0	
537	480.0	0.0	0.0	192.0	0.0	
538	480.0	0.0	0.0	192.0	0.0	
539	480.0	0.0	0.0	192.0	0.0	
540	480.0	0.0	0.0	192.0	0.0	
541	333.0	0.0	0.0	192.0	0.0	
542	255.0	0.0	0.0	192.0	0.0	
543	255.0	0.0	0.0	192.0	0.0	
544	289.0	0.0	0.0	192.0	0.0	
545	255.0	0.0	0.0	192.0	0.0	
546	333.0	0.0	0.0	192.0	0.0	
547	333.0	0.0	0.0	192.0	0.0	
548	289.0	0.0	0.0	192.0	0.0	
549	333.0	0.0	0.0	192.0	0.0	
550	393.0	0.0	0.0	192.0	0.0	
551	255.0	0.0	0.0	192.0	0.0	
552	158.8	238.2	0.0	185.7	0.0	1
553	239.6	359.4	0.0	185.7	0.0	
554	238.2	158.8	0.0	185.7	0.0	1
555	181.9	272.8	0.0	185.7	0.0	1
556	193.5	290.2	0.0	185.7	0.0	
557	255.5	170.3	0.0	185.7	0.0	1

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
558	272.8	181.9	0.0	185.7	0.0	1
559	239.6	359.4	0.0	185.7	0.0	
560	220.8	147.2	0.0	185.7	0.0	1
561	397.0	0.0	0.0	185.7	0.0	1
562	382.5	0.0	0.0	185.7	0.0	1
563	210.7	316.1	0.0	185.7	0.0	
564	158.8	238.2	0.0	185.7	0.0	1
565	295.8	0.0	0.0	185.7	0.0	1
566	255.5	170.3	0.0	185.7	0.0	1
567	203.5	135.7	0.0	185.7	0.0	1
568	397.0	0.0	0.0	185.7	0.0	1
569	381.4	0.0	0.0	185.7	0.0	1
570	295.8	0.0	0.0	185.7	0.0	1
571	228.0	342.1	0.0	185.7	0.0	
572	220.8	147.2	0.0	185.7	0.0	1
573	316.1	210.7	0.0	185.7	0.0	
574	135.7	203.5	0.0	185.7	0.0	1
575	238.1	0.0	0.0	185.7	0.0	1
576	339.2	0.0	0.0	185.7	0.0	1
577	135.7	203.5	0.0	185.7	0.0	1
578	193.5	290.2	0.0	185.7	0.0	
579	203.5	135.7	0.0	185.7	0.0	1
580	290.2	193.5	0.0	185.7	0.0	
581	181.9	272.8	0.0	185.7	0.0	1
582	170.3	155.5	0.0	185.7	0.0	1
583	210.7	316.1	0.0	185.7	0.0	
584	228.0	342.1	0.0	185.7	0.0	
585	290.2	193.5	0.0	185.7	0.0	
586	381.4	0.0	0.0	185.7	0.0	1
587	238.2	158.8	0.0	185.7	0.0	1
588	186.2	124.1	0.0	185.7	0.0	1

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_agg
589	339.2	0.0	0.0	185.7	0.0	1
590	238.1	0.0	0.0	185.7	0.0	1
591	252.5	0.0	0.0	185.7	0.0	1
592	382.5	0.0	0.0	185.7	0.0	1
593	252.5	0.0	0.0	185.7	0.0	1
594	316.1	210.7	0.0	185.7	0.0	
595	186.2	124.1	0.0	185.7	0.0	1
596	170.3	155.5	0.0	185.7	0.0	1
597	272.8	181.9	0.0	185.7	0.0	1
598	339.0	0.0	0.0	197.0	0.0	
599	339.0	0.0	0.0	197.0	0.0	
600	339.0	0.0	0.0	197.0	0.0	
601	339.0	0.0	0.0	197.0	0.0	
602	339.0	0.0	0.0	197.0	0.0	
603	339.0	0.0	0.0	197.0	0.0	
604	339.0	0.0	0.0	197.0	0.0	
605	236.0	0.0	0.0	194.0	0.0	
606	236.0	0.0	0.0	194.0	0.0	
607	236.0	0.0	0.0	194.0	0.0	
608	236.0	0.0	0.0	194.0	0.0	
609	236.0	0.0	0.0	193.0	0.0	
610	236.0	0.0	0.0	193.0	0.0	
611	277.0	0.0	0.0	191.0	0.0	
612	277.0	0.0	0.0	191.0	0.0	
613	277.0	0.0	0.0	191.0	0.0	
614	277.0	0.0	0.0	191.0	0.0	
615	277.0	0.0	0.0	191.0	0.0	
616	277.0	0.0	0.0	191.0	0.0	
617	254.0	0.0	0.0	198.0	0.0	
618	254.0	0.0	0.0	198.0	0.0	
619	254.0	0.0	0.0	198.0	0.0	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_agg
620	254.0	0.0	0.0	198.0	0.0	
621	307.0	0.0	0.0	193.0	0.0	
622	307.0	0.0	0.0	193.0	0.0	
623	307.0	0.0	0.0	193.0	0.0	
624	307.0	0.0	0.0	193.0	0.0	
625	307.0	0.0	0.0	193.0	0.0	
626	236.0	0.0	0.0	193.0	0.0	
627	200.0	0.0	0.0	180.0	0.0	1
628	200.0	0.0	0.0	180.0	0.0	1
629	225.0	0.0	0.0	181.0	0.0	1
630	225.0	0.0	0.0	181.0	0.0	1
631	325.0	0.0	0.0	184.0	0.0	1
632	325.0	0.0	0.0	184.0	0.0	1
633	275.0	0.0	0.0	183.0	0.0	1
634	275.0	0.0	0.0	183.0	0.0	1
635	300.0	0.0	0.0	184.0	0.0	1
636	300.0	0.0	0.0	184.0	0.0	1
637	375.0	0.0	0.0	186.0	0.0	1
638	375.0	0.0	0.0	186.0	0.0	1
639	400.0	0.0	0.0	187.0	0.0	1
640	400.0	0.0	0.0	187.0	0.0	1
641	250.0	0.0	0.0	182.0	0.0	1
642	250.0	0.0	0.0	182.0	0.0	1
643	350.0	0.0	0.0	186.0	0.0	1
644	350.0	0.0	0.0	186.0	0.0	1
645	203.5	305.3	0.0	203.5	0.0	
646	250.2	166.8	0.0	203.5	0.0	
647	157.0	236.0	0.0	192.0	0.0	
648	141.3	212.0	0.0	203.5	0.0	
649	166.8	250.2	0.0	203.5	0.0	
650	122.6	183.9	0.0	203.5	0.0	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
651	183.9	122.6	0.0	203.5	0.0	
652	102.0	153.0	0.0	192.0	0.0	
653	102.0	153.0	0.0	192.0	0.0	
654	122.6	183.9	0.0	203.5	0.0	
655	166.8	250.2	0.0	203.5	0.0	
656	200.0	133.0	0.0	192.0	0.0	
657	108.3	162.4	0.0	203.5	0.0	
658	305.3	203.5	0.0	203.5	0.0	
659	108.3	162.4	0.0	203.5	0.0	
660	116.0	173.0	0.0	192.0	0.0	
661	141.3	212.0	0.0	203.5	0.0	
662	157.0	236.0	0.0	192.0	0.0	
663	133.0	200.0	0.0	192.0	0.0	
664	250.2	166.8	0.0	203.5	0.0	
665	173.0	116.0	0.0	192.0	0.0	
666	192.0	288.0	0.0	192.0	0.0	
667	192.0	288.0	0.0	192.0	0.0	
668	153.0	102.0	0.0	192.0	0.0	
669	288.0	192.0	0.0	192.0	0.0	
670	305.3	203.5	0.0	203.5	0.0	
671	236.0	157.0	0.0	192.0	0.0	
672	173.0	116.0	0.0	192.0	0.0	
673	212.0	141.3	0.0	203.5	0.0	
674	236.0	157.0	0.0	192.0	0.0	
675	183.9	122.6	0.0	203.5	0.0	
676	166.8	250.2	0.0	203.5	0.0	
677	102.0	153.0	0.0	192.0	0.0	
678	288.0	192.0	0.0	192.0	0.0	
679	212.0	141.3	0.0	203.5	0.0	
680	102.0	153.0	0.0	192.0	0.0	
681	173.0	116.0	0.0	192.0	0.0	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
682	183.9	122.6	0.0	203.5	0.0	
683	133.0	200.0	0.0	192.0	0.0	
684	192.0	288.0	0.0	192.0	0.0	
685	133.0	200.0	0.0	192.0	0.0	
686	305.3	203.5	0.0	203.5	0.0	
687	236.0	157.0	0.0	192.0	0.0	
688	108.3	162.4	0.0	203.5	0.0	
689	157.0	236.0	0.0	192.0	0.0	
690	288.0	192.0	0.0	192.0	0.0	
691	212.0	141.3	0.0	203.5	0.0	
692	212.0	141.3	0.0	203.5	0.0	
693	153.0	102.0	0.0	192.0	0.0	
694	236.0	157.0	0.0	192.0	0.0	
695	116.0	173.0	0.0	192.0	0.0	
696	183.9	122.6	0.0	203.5	0.0	
697	108.3	162.4	0.0	203.5	0.0	
698	203.5	305.3	0.0	203.5	0.0	
699	203.5	305.3	0.0	203.5	0.0	
700	133.0	200.0	0.0	192.0	0.0	
701	288.0	192.0	0.0	192.0	0.0	
702	200.0	133.0	0.0	192.0	0.0	
703	200.0	133.0	0.0	192.0	0.0	
704	250.2	166.8	0.0	203.5	0.0	
705	122.6	183.9	0.0	203.5	0.0	
706	153.0	102.0	0.0	192.0	0.0	
707	200.0	133.0	0.0	192.0	0.0	
708	116.0	173.0	0.0	192.0	0.0	
709	173.0	116.0	0.0	192.0	0.0	
710	250.2	166.8	0.0	203.5	0.0	
711	305.3	203.5	0.0	203.5	0.0	
712	192.0	288.0	0.0	192.0	0.0	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_agg
713	157.0	236.0	0.0	192.0	0.0	
714	153.0	102.0	0.0	192.0	0.0	
715	141.3	212.0	0.0	203.5	0.0	
716	116.0	173.0	0.0	192.0	0.0	
717	141.3	212.0	0.0	203.5	0.0	
718	122.6	183.9	0.0	203.5	0.0	
719	166.8	250.2	0.0	203.5	0.0	
720	203.5	305.3	0.0	203.5	0.0	
721	310.0	0.0	0.0	192.0	0.0	1
722	310.0	0.0	0.0	192.0	0.0	1
723	310.0	0.0	0.0	192.0	0.0	1
724	310.0	0.0	0.0	192.0	0.0	1
725	310.0	0.0	0.0	192.0	0.0	1
726	331.0	0.0	0.0	192.0	0.0	1
727	331.0	0.0	0.0	192.0	0.0	1
728	331.0	0.0	0.0	192.0	0.0	1
729	331.0	0.0	0.0	192.0	0.0	1
730	331.0	0.0	0.0	192.0	0.0	1
731	349.0	0.0	0.0	192.0	0.0	1
732	349.0	0.0	0.0	192.0	0.0	1
733	349.0	0.0	0.0	192.0	0.0	1
734	349.0	0.0	0.0	192.0	0.0	1
735	349.0	0.0	0.0	192.0	0.0	1
736	238.0	0.0	0.0	186.0	0.0	1
737	238.0	0.0	0.0	186.0	0.0	1
738	296.0	0.0	0.0	186.0	0.0	1
739	296.0	0.0	0.0	186.0	0.0	1
740	297.0	0.0	0.0	186.0	0.0	1
741	480.0	0.0	0.0	192.0	0.0	
742	480.0	0.0	0.0	192.0	0.0	
743	397.0	0.0	0.0	186.0	0.0	1

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_agg
744	281.0	0.0	0.0	186.0	0.0	1
745	281.0	0.0	0.0	185.0	0.0	1
746	500.0	0.0	0.0	200.0	0.0	1
747	500.0	0.0	0.0	200.0	0.0	1
748	500.0	0.0	0.0	200.0	0.0	1
749	500.0	0.0	0.0	200.0	0.0	1
750	500.0	0.0	0.0	200.0	0.0	1
751	540.0	0.0	0.0	173.0	0.0	1
752	540.0	0.0	0.0	173.0	0.0	1
753	540.0	0.0	0.0	173.0	0.0	1
754	540.0	0.0	0.0	173.0	0.0	1
755	540.0	0.0	0.0	173.0	0.0	1
756	540.0	0.0	0.0	173.0	0.0	1
757	350.0	0.0	0.0	203.0	0.0	
758	350.0	0.0	0.0	203.0	0.0	
759	350.0	0.0	0.0	203.0	0.0	
760	350.0	0.0	0.0	203.0	0.0	
761	350.0	0.0	0.0	203.0	0.0	
762	350.0	0.0	0.0	203.0	0.0	
763	385.0	0.0	0.0	186.0	0.0	
764	385.0	0.0	0.0	186.0	0.0	
765	385.0	0.0	0.0	186.0	0.0	
766	385.0	0.0	0.0	186.0	0.0	
767	385.0	0.0	0.0	186.0	0.0	
768	331.0	0.0	0.0	192.0	0.0	
769	331.0	0.0	0.0	192.0	0.0	
770	349.0	0.0	0.0	192.0	0.0	1
771	331.0	0.0	0.0	192.0	0.0	
772	382.0	0.0	0.0	186.0	0.0	1
773	382.0	0.0	0.0	186.0	0.0	1
774	382.0	0.0	0.0	186.0	0.0	1

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_agg
775	281.0	0.0	0.0	186.0	0.0	1
776	339.0	0.0	0.0	185.0	0.0	1
777	339.0	0.0	0.0	185.0	0.0	1
778	295.0	0.0	0.0	185.0	0.0	1
779	295.0	0.0	0.0	185.0	0.0	1
780	238.0	0.0	0.0	185.0	0.0	1
781	296.0	0.0	0.0	192.0	0.0	1
782	296.0	0.0	0.0	192.0	0.0	1
783	296.0	0.0	0.0	192.0	0.0	1
784	331.0	0.0	0.0	192.0	0.0	
785	331.0	0.0	0.0	192.0	0.0	
786	331.0	0.0	0.0	192.0	0.0	
787	331.0	0.0	0.0	192.0	0.0	
788	349.0	0.0	0.0	192.0	0.0	1
789	349.0	0.0	0.0	192.0	0.0	1
790	349.0	0.0	0.0	192.0	0.0	1
791	349.0	0.0	0.0	192.0	0.0	1
792	349.0	0.0	0.0	192.0	0.0	1
793	302.0	0.0	0.0	203.0	0.0	
794	302.0	0.0	0.0	203.0	0.0	
795	525.0	0.0	0.0	189.0	0.0	1
796	500.0	0.0	0.0	200.0	0.0	1
797	500.0	0.0	0.0	200.0	0.0	1
798	500.0	0.0	0.0	200.0	0.0	1
799	540.0	0.0	0.0	173.0	0.0	1
800	252.0	0.0	0.0	185.0	0.0	1
801	252.0	0.0	0.0	185.0	0.0	1
802	339.0	0.0	0.0	185.0	0.0	1
803	393.0	0.0	0.0	192.0	0.0	
804	393.0	0.0	0.0	192.0	0.0	
805	393.0	0.0	0.0	192.0	0.0	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_agg
806	382.0	0.0	0.0	185.0	0.0	1
807	382.0	0.0	0.0	185.0	0.0	1
808	252.0	0.0	0.0	186.0	0.0	1
809	252.0	0.0	0.0	185.0	0.0	1
810	310.0	0.0	0.0	192.0	0.0	
811	310.0	0.0	0.0	192.0	0.0	
812	310.0	0.0	0.0	192.0	0.0	
813	310.0	0.0	0.0	192.0	0.0	
814	310.0	0.0	0.0	192.0	0.0	
815	525.0	0.0	0.0	189.0	0.0	1
816	525.0	0.0	0.0	189.0	0.0	1
817	525.0	0.0	0.0	189.0	0.0	1
818	525.0	0.0	0.0	189.0	0.0	1
819	525.0	0.0	0.0	189.0	0.0	1
820	525.0	0.0	0.0	189.0	0.0	1
821	322.0	0.0	0.0	203.0	0.0	
822	322.0	0.0	0.0	203.0	0.0	
823	322.0	0.0	0.0	203.0	0.0	
824	302.0	0.0	0.0	203.0	0.0	
825	397.0	0.0	0.0	185.0	0.0	1
826	480.0	0.0	0.0	192.0	0.0	
827	522.0	0.0	0.0	146.0	0.0	
828	522.0	0.0	0.0	146.0	0.0	
829	273.0	105.0	82.0	210.0	9.0	
830	162.0	190.0	148.0	179.0	19.0	
831	154.0	144.0	112.0	220.0	10.0	
832	147.0	115.0	89.0	202.0	9.0	
833	152.0	178.0	139.0	168.0	18.0	
834	310.0	143.0	111.0	168.0	22.0	
835	144.0	0.0	175.0	158.0	18.0	
836	304.0	140.0	0.0	214.0	6.0	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
837	374.0	0.0	0.0	190.0	7.0	1
838	159.0	149.0	116.0	175.0	15.0	
839	153.0	239.0	0.0	200.0	6.0	1
840	310.0	143.0	0.0	168.0	10.0	
841	305.0	0.0	100.0	196.0	10.0	
842	151.0	0.0	184.0	167.0	12.0	
843	142.0	167.0	130.0	174.0	11.0	
844	298.0	137.0	107.0	201.0	6.0	
845	321.0	164.0	0.0	190.0	5.0	
846	366.0	187.0	0.0	191.0	7.0	
847	280.0	129.0	100.0	172.0	9.0	
848	252.0	97.0	76.0	194.0	8.0	
849	165.0	0.0	150.0	182.0	12.0	1
850	156.0	243.0	0.0	180.0	11.0	1
851	160.0	188.0	146.0	203.0	11.0	
852	298.0	0.0	107.0	186.0	6.0	
853	318.0	0.0	126.0	210.0	6.0	
854	287.0	121.0	94.0	188.0	9.0	
855	326.0	166.0	0.0	174.0	9.0	
856	356.0	0.0	142.0	193.0	11.0	
857	132.0	207.0	161.0	179.0	5.0	
858	322.0	149.0	0.0	186.0	8.0	
859	164.0	0.0	200.0	181.0	13.0	
860	314.0	0.0	113.0	170.0	10.0	
861	321.0	0.0	128.0	182.0	11.0	
862	140.0	164.0	128.0	237.0	6.0	
863	288.0	121.0	0.0	177.0	7.0	
864	298.0	0.0	107.0	210.0	11.0	
865	265.0	111.0	86.0	195.0	6.0	
866	160.0	250.0	0.0	168.0	12.0	1
867	166.0	260.0	0.0	183.0	13.0	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
868	276.0	116.0	90.0	180.0	9.0	
869	322.0	0.0	116.0	196.0	10.0	
870	149.0	139.0	109.0	193.0	6.0	
871	159.0	187.0	0.0	176.0	11.0	
872	261.0	100.0	78.0	201.0	9.0	
873	237.0	92.0	71.0	247.0	6.0	
874	313.0	0.0	113.0	178.0	8.0	1
875	155.0	183.0	0.0	193.0	9.0	1
876	146.0	230.0	0.0	202.0	3.0	
877	296.0	0.0	107.0	221.0	11.0	
878	133.0	210.0	0.0	196.0	3.0	
879	313.0	145.0	0.0	178.0	8.0	
880	152.0	0.0	112.0	184.0	8.0	
881	153.0	145.0	113.0	178.0	8.0	1
882	140.0	133.0	103.0	200.0	7.0	
883	149.0	236.0	0.0	176.0	13.0	
884	300.0	0.0	120.0	212.0	10.0	
885	153.0	145.0	113.0	178.0	8.0	
886	148.0	0.0	137.0	158.0	16.0	1
887	326.0	0.0	138.0	199.0	11.0	
888	153.0	145.0	0.0	178.0	8.0	1
889	262.0	111.0	86.0	195.0	5.0	
890	158.0	0.0	195.0	220.0	11.0	
891	151.0	0.0	185.0	167.0	16.0	1
892	273.0	0.0	90.0	199.0	11.0	
893	149.0	118.0	92.0	183.0	7.0	
894	143.0	169.0	143.0	191.0	8.0	
895	260.0	101.0	78.0	171.0	10.0	
896	313.0	161.0	0.0	178.0	10.0	
897	284.0	120.0	0.0	168.0	7.0	
898	336.0	0.0	0.0	182.0	3.0	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggreg
899	145.0	0.0	134.0	181.0	11.0	
900	150.0	237.0	0.0	174.0	12.0	1
901	144.0	170.0	133.0	192.0	8.0	
902	331.0	170.0	0.0	195.0	8.0	
903	155.0	0.0	143.0	193.0	9.0	1
904	155.0	183.0	0.0	193.0	9.0	
905	135.0	0.0	166.0	180.0	10.0	
906	266.0	112.0	87.0	178.0	10.0	
907	314.0	145.0	113.0	179.0	8.0	
908	313.0	145.0	0.0	127.0	8.0	1
909	146.0	173.0	0.0	182.0	3.0	
910	144.0	136.0	106.0	178.0	7.0	
911	148.0	0.0	182.0	181.0	15.0	
912	277.0	117.0	91.0	191.0	7.0	
913	298.0	0.0	107.0	164.0	13.0	
914	313.0	145.0	0.0	178.0	8.0	1
915	155.0	184.0	143.0	194.0	9.0	
916	289.0	134.0	0.0	195.0	6.0	
917	148.0	175.0	0.0	171.0	2.0	1
918	145.0	0.0	179.0	202.0	8.0	
919	313.0	0.0	0.0	178.0	8.0	1
920	136.0	162.0	126.0	172.0	10.0	
921	155.0	0.0	143.0	193.0	9.0	
922	255.0	99.0	77.0	189.0	6.0	
923	162.0	207.0	172.0	216.0	10.0	
924	136.0	196.0	98.0	199.0	6.0	
925	164.0	163.0	128.0	197.0	8.0	
926	162.0	214.0	164.0	202.0	10.0	
927	157.0	214.0	152.0	200.0	9.0	
928	149.0	153.0	194.0	192.0	8.0	
929	135.0	105.0	193.0	196.0	6.0	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggre
930	159.0	209.0	161.0	201.0	7.0	
931	144.0	15.0	195.0	176.0	6.0	1
932	154.0	174.0	185.0	228.0	7.0	
933	167.0	187.0	195.0	185.0	7.0	
934	184.0	86.0	190.0	213.0	6.0	
935	156.0	178.0	187.0	221.0	7.0	
936	236.9	91.7	71.5	246.9	6.0	
937	313.3	0.0	113.0	178.5	8.0	1
938	154.8	183.4	0.0	193.3	9.1	1
939	145.9	230.5	0.0	202.5	3.4	
940	296.0	0.0	106.7	221.4	10.5	
941	133.1	210.2	0.0	195.7	3.1	
942	313.3	145.0	0.0	178.5	8.0	
943	151.6	0.0	111.9	184.4	7.9	
944	153.1	145.0	113.0	178.5	8.0	1
945	139.9	132.6	103.3	200.3	7.4	
946	149.5	236.0	0.0	175.8	12.6	
947	299.8	0.0	119.8	211.5	9.9	
948	153.1	145.0	113.0	178.5	8.0	
949	148.1	0.0	136.6	158.1	16.1	1
950	326.5	0.0	137.9	199.0	10.8	
951	152.7	144.7	0.0	178.1	8.0	
952	261.9	110.5	86.1	195.4	5.0	
953	158.4	0.0	194.9	219.7	11.0	
954	150.7	0.0	185.3	166.7	15.6	1
955	272.6	0.0	89.6	198.7	10.6	
956	149.0	117.6	91.7	182.9	7.1	
957	143.0	169.4	142.7	190.7	8.4	
958	259.9	100.6	78.4	170.6	10.4	
959	312.9	160.5	0.0	177.6	9.6	
960	284.0	119.7	0.0	168.3	7.2	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_agg
961	336.5	0.0	0.0	181.9	3.4	
962	144.8	0.0	133.6	180.8	11.1	
963	150.0	236.8	0.0	173.8	11.9	1
964	143.7	170.2	132.6	191.6	8.5	
965	330.5	169.6	0.0	194.9	8.1	
966	154.8	0.0	142.8	193.3	9.1	1
967	154.8	183.4	0.0	193.3	9.1	
968	134.7	0.0	165.7	180.2	10.0	
969	266.2	112.3	87.5	177.9	10.4	
970	314.0	145.3	113.2	178.9	8.0	
971	312.7	144.7	0.0	127.3	8.0	
972	145.7	172.6	0.0	181.9	3.4	
973	143.8	136.3	106.2	178.1	7.5	
974	148.1	0.0	182.1	181.4	15.0	
975	277.0	116.8	91.0	190.6	7.0	
976	298.1	0.0	107.5	163.6	12.8	
977	313.3	145.0	0.0	178.5	8.0	1
978	155.2	183.9	143.2	193.8	9.2	
979	289.0	133.7	0.0	194.9	5.5	
980	147.8	175.1	0.0	171.2	2.2	1
981	145.4	0.0	178.9	201.7	7.8	
982	312.7	0.0	0.0	178.1	8.0	
983	136.4	161.6	125.8	171.6	10.4	
984	154.8	0.0	142.8	193.3	9.1	
985	255.3	98.8	77.0	188.6	6.5	
986	272.8	105.1	81.8	209.7	9.0	
987	162.0	190.1	148.1	178.8	18.8	
988	153.6	144.2	112.3	220.1	10.1	
989	146.5	114.6	89.3	201.9	8.8	
990	151.8	178.1	138.7	167.5	18.3	
991	309.9	142.8	111.2	167.8	22.1	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_agg
992	143.6	0.0	174.9	158.4	17.9	
993	303.6	139.9	0.0	213.5	6.2	
994	374.3	0.0	0.0	190.2	6.7	1
995	158.6	148.9	116.0	175.1	15.0	
996	152.6	238.7	0.0	200.0	6.3	1
997	310.0	142.8	0.0	167.9	10.0	
998	304.8	0.0	99.6	196.0	9.8	
999	150.9	0.0	183.9	166.6	11.6	
1000	141.9	166.6	129.7	173.5	10.9	
1001	297.8	137.2	106.9	201.3	6.0	
1002	321.3	164.2	0.0	190.5	4.6	
1003	366.0	187.0	0.0	191.3	6.6	
1004	279.8	128.9	100.4	172.4	9.5	
1005	252.1	97.1	75.6	193.8	8.3	
1006	164.6	0.0	150.4	181.6	11.7	1
1007	155.6	243.5	0.0	180.3	10.7	1
1008	160.2	188.0	146.4	203.2	11.3	
1009	298.1	0.0	107.0	186.4	6.1	
1010	317.9	0.0	126.5	209.7	5.7	
1011	287.3	120.5	93.9	187.6	9.2	
1012	325.6	166.4	0.0	174.0	8.9	
1013	355.9	0.0	141.6	193.3	11.0	
1014	132.0	206.5	160.9	178.9	5.5	
1015	322.5	148.6	0.0	185.8	8.5	
1016	164.2	0.0	200.1	181.2	12.6	
1017	313.8	0.0	112.6	169.9	10.1	
1018	321.4	0.0	127.9	182.5	11.5	
1019	139.7	163.9	127.7	236.7	5.8	
1020	288.4	121.0	0.0	177.4	7.0	
1021	298.2	0.0	107.0	209.7	11.1	
1022	264.5	111.0	86.5	195.5	5.9	

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggreg
1023	159.8	250.0	0.0	168.4	12.2	1
1024	166.0	259.7	0.0	183.2	12.7	
1025	276.4	116.0	90.3	179.6	8.9	
1026	322.2	0.0	115.6	196.0	10.4	
1027	148.5	139.4	108.6	192.7	6.1	
1028	159.1	186.7	0.0	175.6	11.3	
1029	260.9	100.5	78.3	200.6	8.6	

```
In [6]: # SHOWING 5 ROWS
df.head()
```

```
Out[6]:
```

	cement	blast_furnace_slag	fly_ash	water	superplasticizer	coarse_aggregat
0	540.0	0.0	0.0	162.0	2.5	1040.
1	540.0	0.0	0.0	162.0	2.5	1055.
2	332.5	142.5	0.0	228.0	0.0	932.
3	332.5	142.5	0.0	228.0	0.0	932.
4	198.6	132.4	0.0	192.0	0.0	978.

```
In [7]: # STEP 4: BASIC DATA CHECK

print("SHAPE:\n", df.shape)
```

```
SHAPE:
(1030, 9)
```

```
In [8]: print("\nINFO:\n")
print(df.info())
```

INFO:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1030 entries, 0 to 1029
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   cement                                1030 non-null   float64
1   blast_furnace_slag                   1030 non-null   float64
2   fly_ash                              1030 non-null   float64
3   water                                1030 non-null   float64
4   superplasticizer                     1030 non-null   float64
5   coarse_aggregate                     1030 non-null   float64
6   fine_aggregate                       1030 non-null   float64
7   age                                  1030 non-null   int64
8   concrete_compressive_strength        1030 non-null   float64
dtypes: float64(8), int64(1)
memory usage: 72.6 KB
None
```

```
In [9]: print("\nDESCRIBE:\n")
        print(df.describe(include='all'))
```

DESCRIBE:

	cement	blast_furnace_slag	fly_ash	water	\
count	1030.000000	1030.000000	1030.000000	1030.000000	
mean	281.167864	73.895825	54.188350	181.567282	
std	104.506364	86.279342	63.997004	21.354219	
min	102.000000	0.000000	0.000000	121.800000	
25%	192.375000	0.000000	0.000000	164.900000	
50%	272.900000	22.000000	0.000000	185.000000	
75%	350.000000	142.950000	118.300000	192.000000	
max	540.000000	359.400000	200.100000	247.000000	

	superplasticizer	coarse_aggregate	fine_aggregate	age	\
count	1030.000000	1030.000000	1030.000000	1030.000000	
mean	6.204660	972.918932	773.580485	45.662136	
std	5.973841	77.753954	80.175980	63.169912	
min	0.000000	801.000000	594.000000	1.000000	
25%	0.000000	932.000000	730.950000	7.000000	
50%	6.400000	968.000000	779.500000	28.000000	
75%	10.200000	1029.400000	824.000000	56.000000	
max	32.200000	1145.000000	992.600000	365.000000	

	concrete_compressive_strength
count	1030.000000
mean	35.817961
std	16.705742
min	2.330000
25%	23.710000
50%	34.445000
75%	46.135000
max	82.600000

```
In [10]: print("\nCOLUMNS:\n")
print(df.columns)
```

COLUMNS:

```
Index(['cement', 'blast_furnace_slag', 'fly_ash', 'water', 'superplasticizer',
      'coarse_aggregate', 'fine_aggregate ', 'age',
      'concrete_compressive_strength'],
      dtype='object')
```

```
In [11]: # STEP 5: CLEAN COLUMN NAMES
df.columns = df.columns.str.strip().str.upper().str.replace(" ", "_")
df.columns
```

```
Out[11]: Index(['CEMENT', 'BLAST_FURNACE_SLAG', 'FLY_ASH', 'WATER', 'SUPERPLASTICIZE
R',
              'COARSE_AGGREGATE', 'FINE_AGGREGATE', 'AGE',
              'CONCRETE_COMPRESSIVE_STRENGTH'],
              dtype='object')
```

```
In [12]: # STEP 6: CHECK MISSING VALUES
df.isnull().sum()      # null values
```

```
Out[12]:
```

	0
CEMENT	0
BLAST_FURNACE_SLAG	0
FLY_ASH	0
WATER	0
SUPERPLASTICIZER	0
COARSE_AGGREGATE	0
FINE_AGGREGATE	0
AGE	0
CONCRETE_COMPRESSIVE_STRENGTH	0

dtype: int64

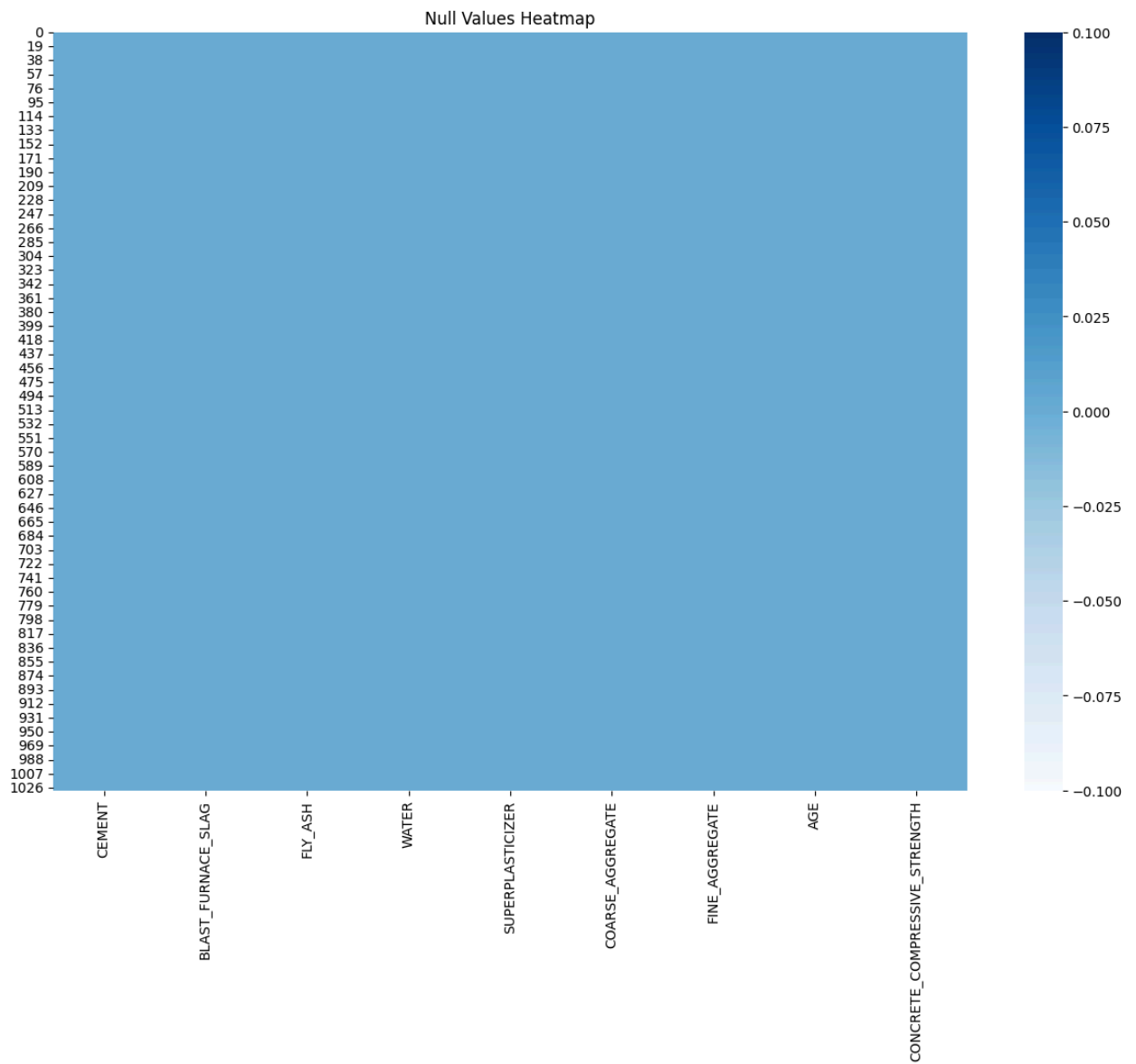
```
In [13]: df.notnull().sum()      # not-null values
```

Out[13]:

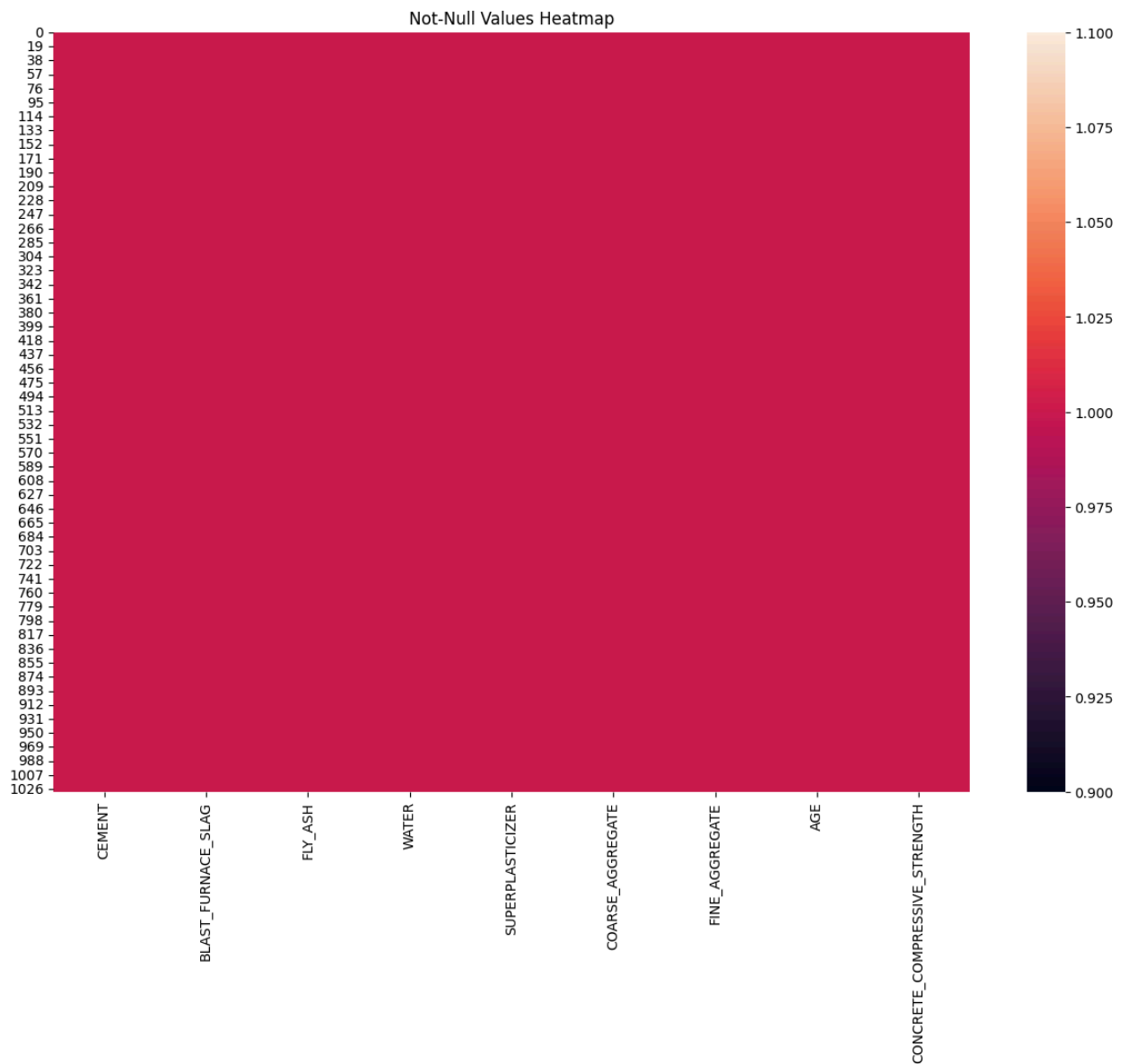
	CEMENT	1030
	BLAST_FURNACE_SLAG	1030
	FLY_ASH	1030
	WATER	1030
	SUPERPLASTICIZER	1030
	COARSE_AGGREGATE	1030
	FINE_AGGREGATE	1030
	AGE	1030
	CONCRETE_COMPRESSIVE_STRENGTH	1030

dtype: int64

```
In [14]: # STEP 7: HEATMAP OF NULL VALUES
plt.figure(figsize=(15, 10))
sns.heatmap(df.isnull(), cmap="Blues", cbar=True)
plt.title("Null Values Heatmap")
plt.show()
```



```
In [15]: plt.figure(figsize=(15, 10))  
sns.heatmap(df.notnull(), cbar=True)  
plt.title("Not-Null Values Heatmap")  
plt.show()
```



```
In [16]: # STEP 8: CHECK DUPLICATES
df.duplicated().sum()
```

```
Out[16]: np.int64(25)
```

```
In [17]: # Drop duplicates
df = df.drop_duplicates()
print("Shape after removing duplicates:", df.shape)
```

Shape after removing duplicates: (1005, 9)

```
In [18]: # STEP 9: CHECK & FIX DATA TYPES
df.dtypes
```

Out[18]:

CEMENT	float64
BLAST_FURNACE_SLAG	float64
FLY_ASH	float64
WATER	float64
SUPERPLASTICIZER	float64
COARSE_AGGREGATE	float64
FINE_AGGREGATE	float64
AGE	int64
CONCRETE_COMPRESSIVE_STRENGTH	float64

dtype: object

```
In [19]: # DROPPING NULL VALUES
df=df.dropna()
```

```
In [20]: # RECHECK
df.head()
```

```
Out[20]:
```

	CEMENT	BLAST_FURNACE_SLAG	FLY_ASH	WATER	SUPERPLASTICIZER	COAR
0	540.0	0.0	0.0	162.0	2.5	
1	540.0	0.0	0.0	162.0	2.5	
2	332.5	142.5	0.0	228.0	0.0	
3	332.5	142.5	0.0	228.0	0.0	
4	198.6	132.4	0.0	192.0	0.0	

```
In [21]: # RECHECK
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 1005 entries, 0 to 1029
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   CEMENT                                1005 non-null   float64
1   BLAST_FURNACE_SLAG                    1005 non-null   float64
2   FLY_ASH                               1005 non-null   float64
3   WATER                                 1005 non-null   float64
4   SUPERPLASTICIZER                      1005 non-null   float64
5   COARSE_AGGREGATE                      1005 non-null   float64
6   FINE_AGGREGATE                        1005 non-null   float64
7   AGE                                   1005 non-null   int64
8   CONCRETE_COMPRESSIVE_STRENGTH         1005 non-null   float64
dtypes: float64(8), int64(1)
memory usage: 78.5 KB
```

```
In [22]: # RECHECK
df.isnull().sum()      # null values
```

```
Out[22]:
```

	0
CEMENT	0
BLAST_FURNACE_SLAG	0
FLY_ASH	0
WATER	0
SUPERPLASTICIZER	0
COARSE_AGGREGATE	0
FINE_AGGREGATE	0
AGE	0
CONCRETE_COMPRESSIVE_STRENGTH	0

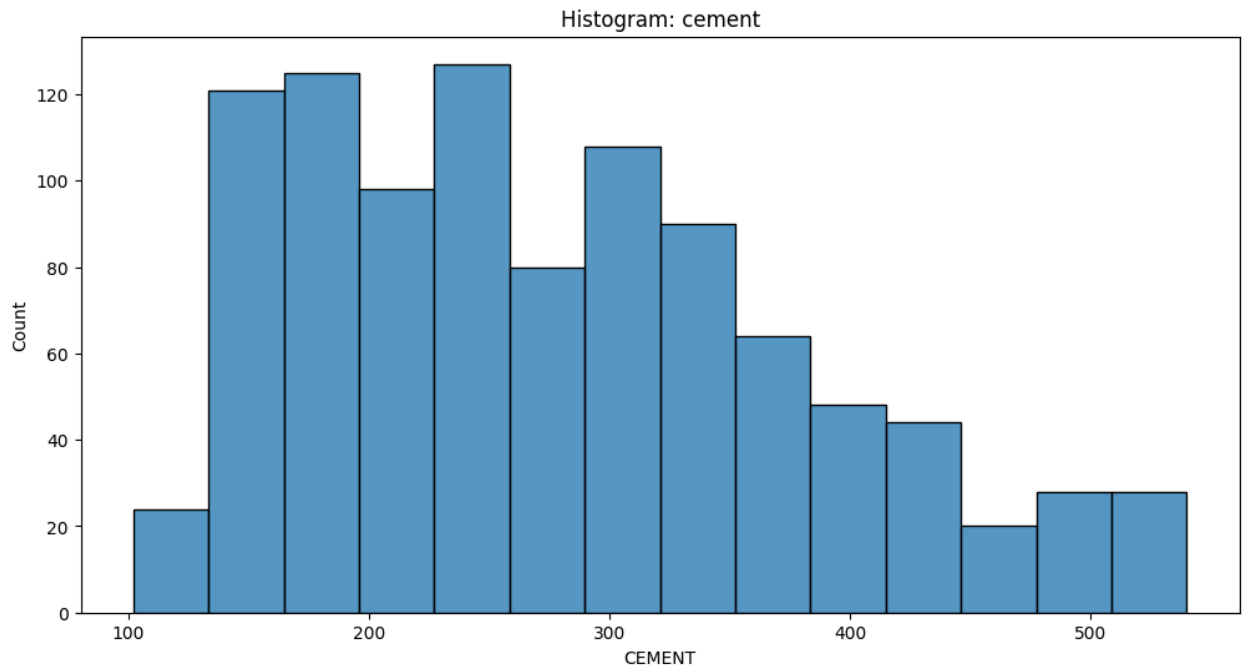
dtype: int64

```
In [23]: df.columns
```

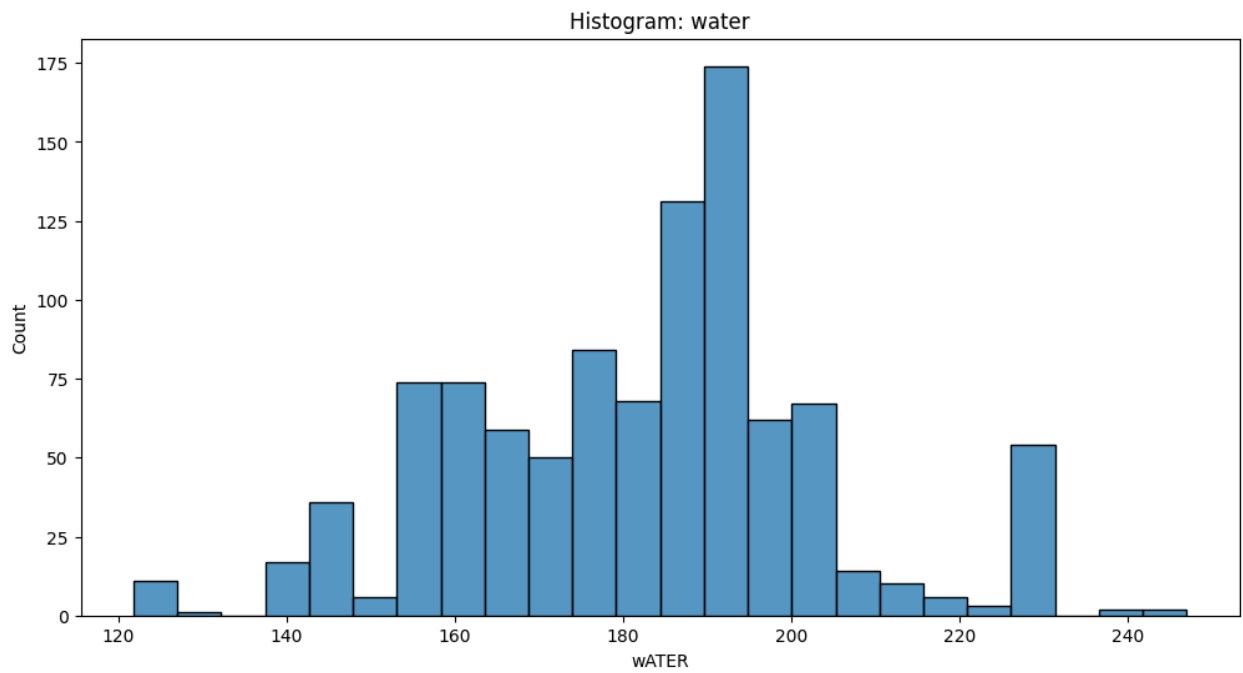
```
Out[23]: Index(['CEMENT', 'BLAST_FURNACE_SLAG', 'FLY_ASH', 'WATER', 'SUPERPLASTICIZER',
               'COARSE_AGGREGATE', 'FINE_AGGREGATE', 'AGE',
               'CONCRETE_COMPRESSIVE_STRENGTH'],
              dtype='object')
```

```
In [24]: # STEP 11: VISUALIZATIONS
# Histogram for cement column
# This shows the distribution & spread of cement values
plt.figure(figsize=(12,6))
sns.histplot(df["CEMENT"])
```

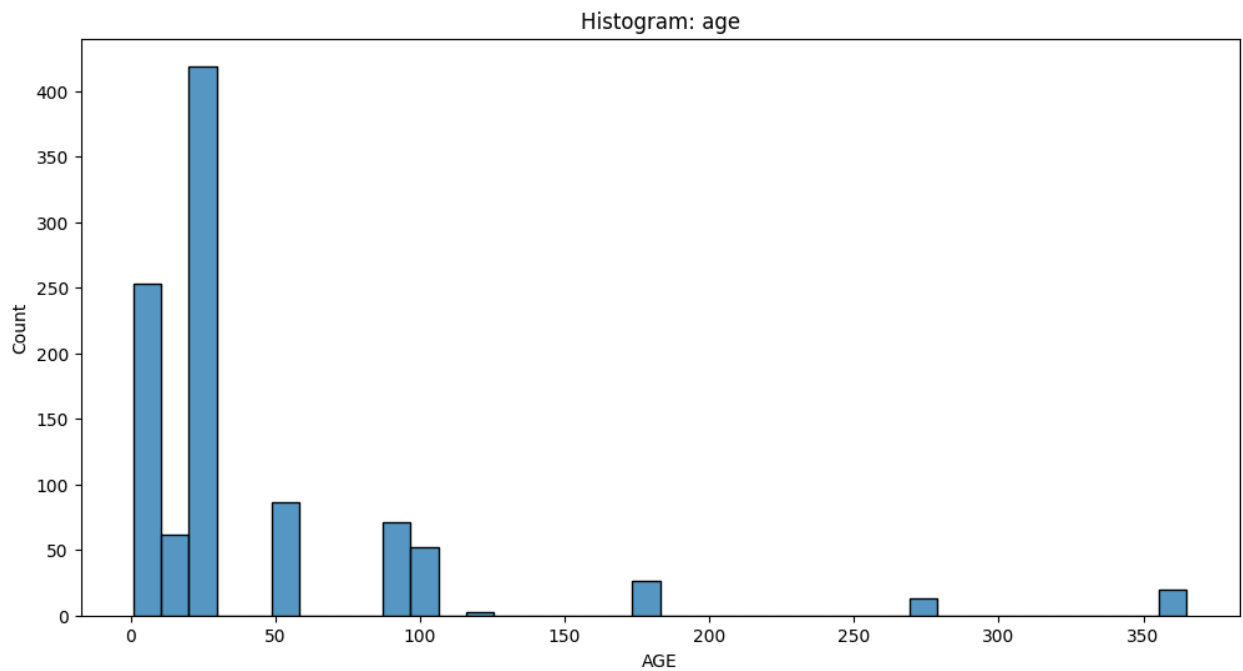
```
plt.xlabel("CEMENT")
plt.ylabel("Count")
plt.title("Histogram: cement")
plt.show()
```



```
In [26]: # Histogram for water
# Helps check if values are normal or skewed
plt.figure(figsize=(12,6))
sns.histplot(df["WATER"])
plt.xlabel("wATER")
plt.ylabel("Count")
plt.title("Histogram: water")
plt.show()
```

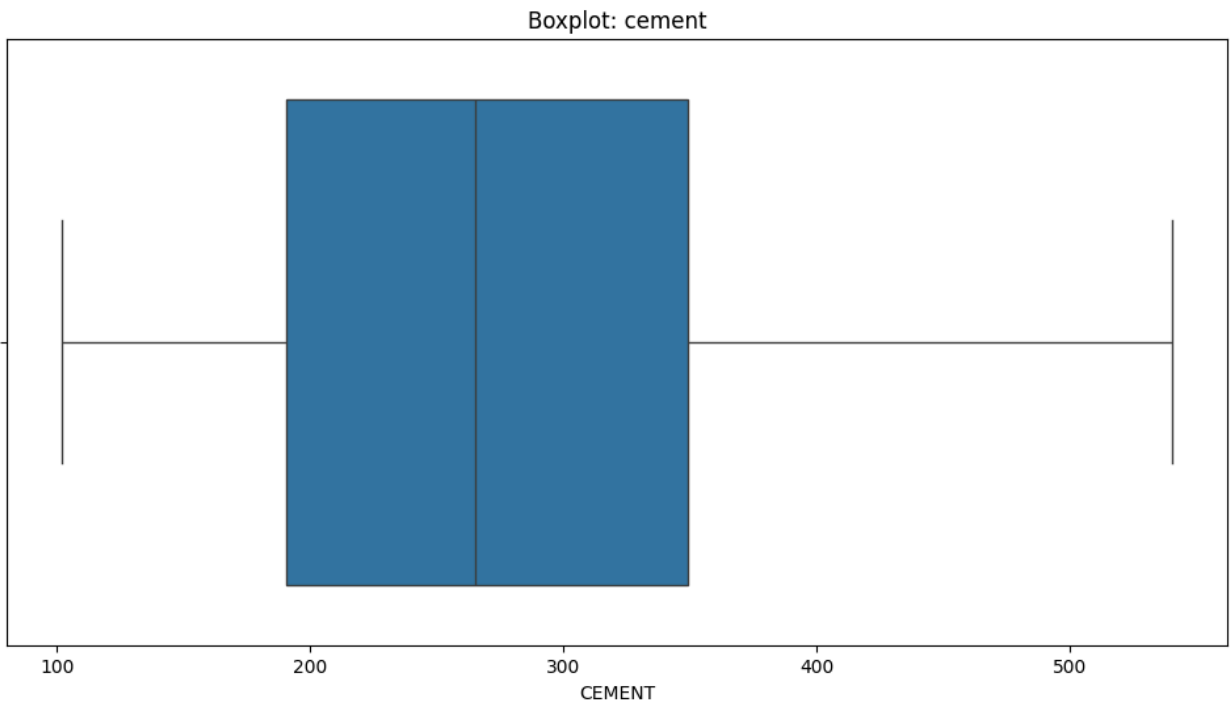


```
In [27]: # Histogram for age
# Shows age spread
plt.figure(figsize=(12,6))
sns.histplot(df["AGE"])
plt.xlabel("AGE")
plt.ylabel("Count")
plt.title("Histogram: age")
plt.show()
```

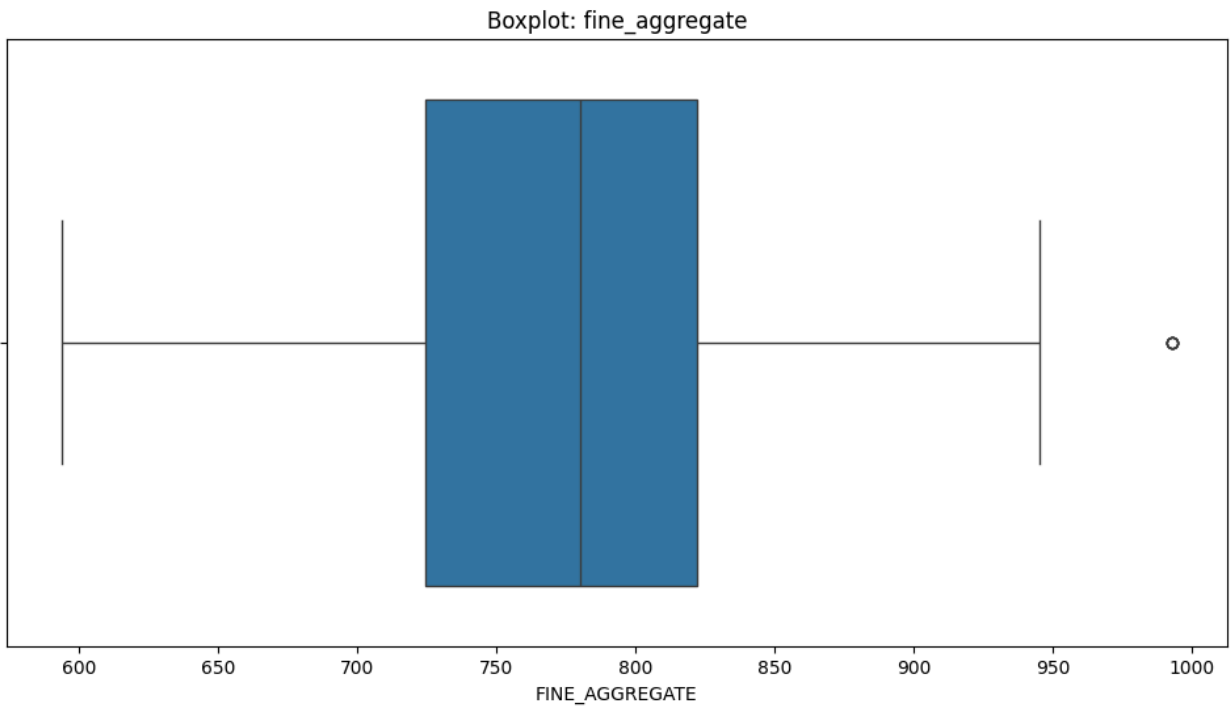


```
In [29]: # Boxplot for cement
# Helps detect outliers in cement values
plt.figure(figsize=(12,6))
```

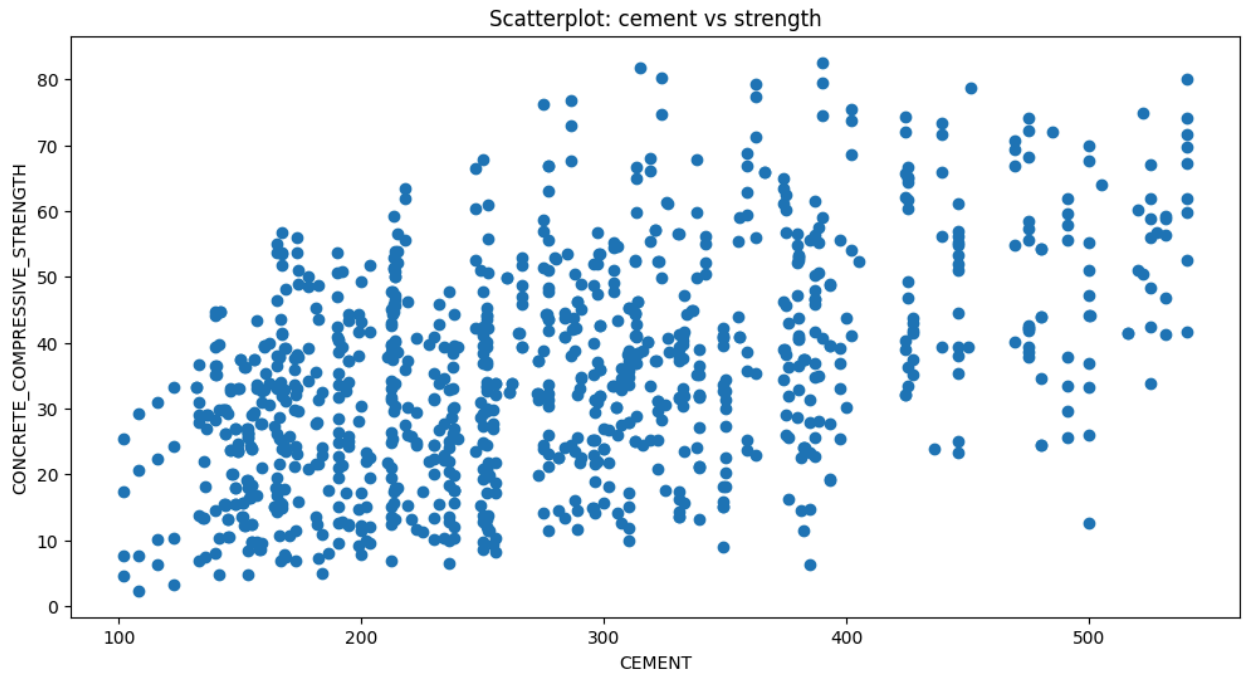
```
sns.boxplot(x=df["CEMENT"])  
plt.title("Boxplot: cement")  
plt.show()
```



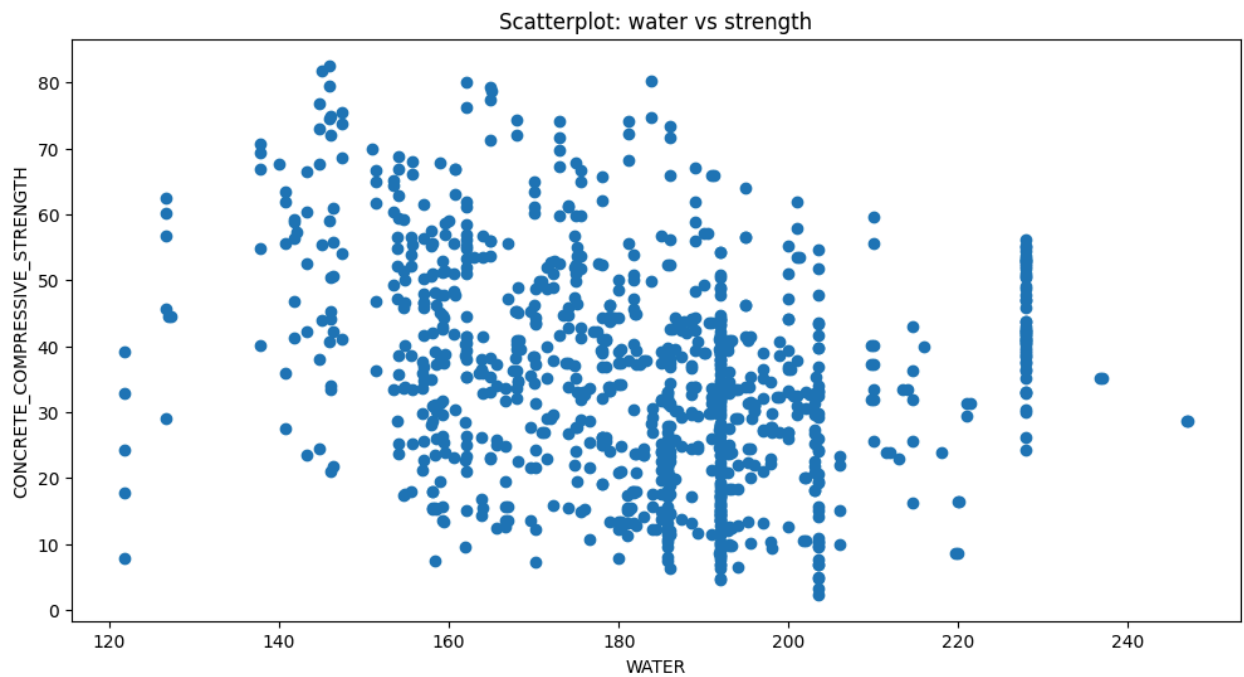
```
In [32]: # Boxplot for fine aggregate  
# Shows distribution and any extreme values  
plt.figure(figsize=(12,6))  
sns.boxplot(x=df["FINE_AGGREGATE"])  
plt.title("Boxplot: fine_aggregate")  
plt.show()
```



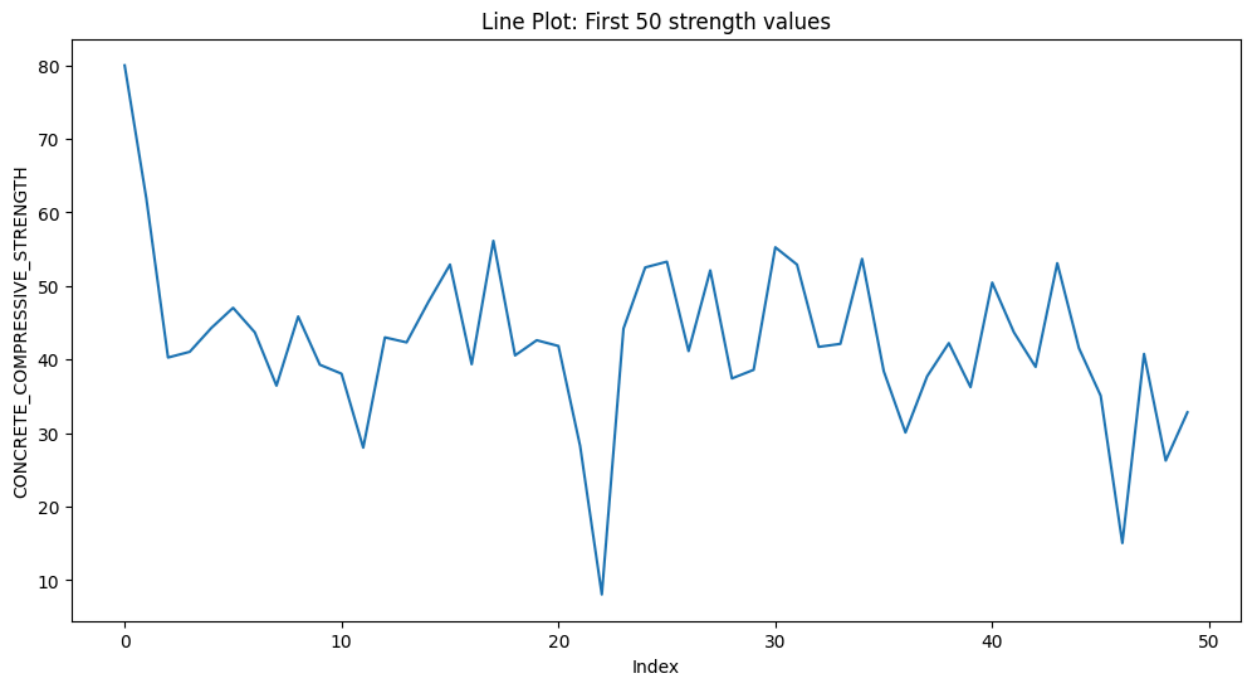
```
In [34]: # Scatterplot: cement vs strength
# Shows relationship: more cement = higher strength
plt.figure(figsize=(12,6))
plt.scatter(df["CEMENT"], df["CONCRETE_COMPRESSIVE_STRENGTH"])
plt.xlabel("CEMENT")
plt.ylabel("CONCRETE_COMPRESSIVE_STRENGTH")
plt.title("Scatterplot: cement vs strength")
plt.show()
```



```
In [35]: # Scatterplot: water vs strength
# Generally, more water reduces strength
plt.figure(figsize=(12,6))
plt.scatter(df["WATER"], df["CONCRETE_COMPRESSIVE_STRENGTH"])
plt.xlabel("WATER")
plt.ylabel("CONCRETE_COMPRESSIVE_STRENGTH")
plt.title("Scatterplot: water vs strength")
plt.show()
```

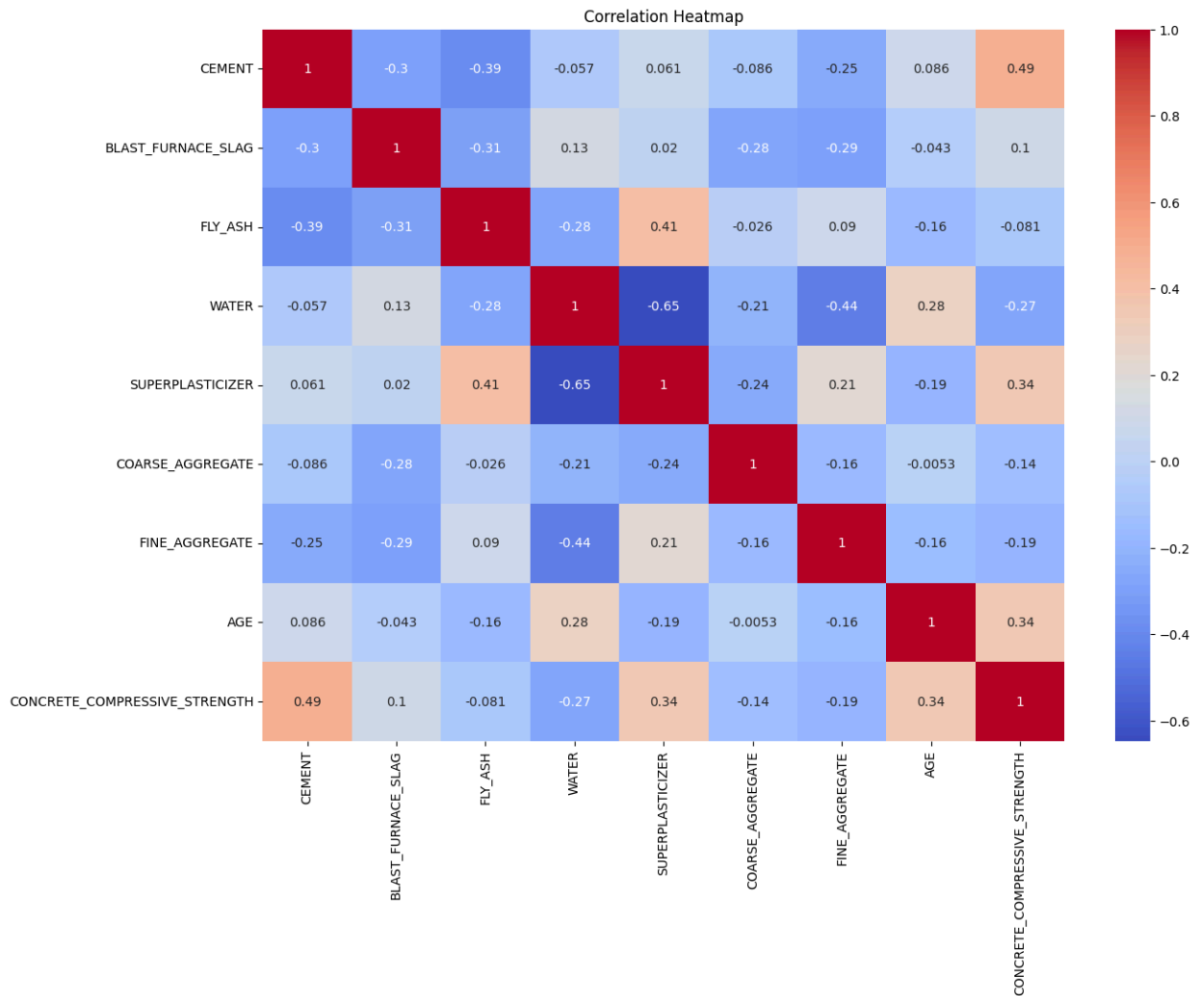


```
In [37]: # Line plot for first 50 strength values
# Helps see small trend patterns
plt.figure(figsize=(12,6))
plt.plot(df["CONCRETE_COMPRESSIVE_STRENGTH"][:50])
plt.xlabel("Index")
plt.ylabel("CONCRETE_COMPRESSIVE_STRENGTH")
plt.title("Line Plot: First 50 strength values")
plt.show()
```

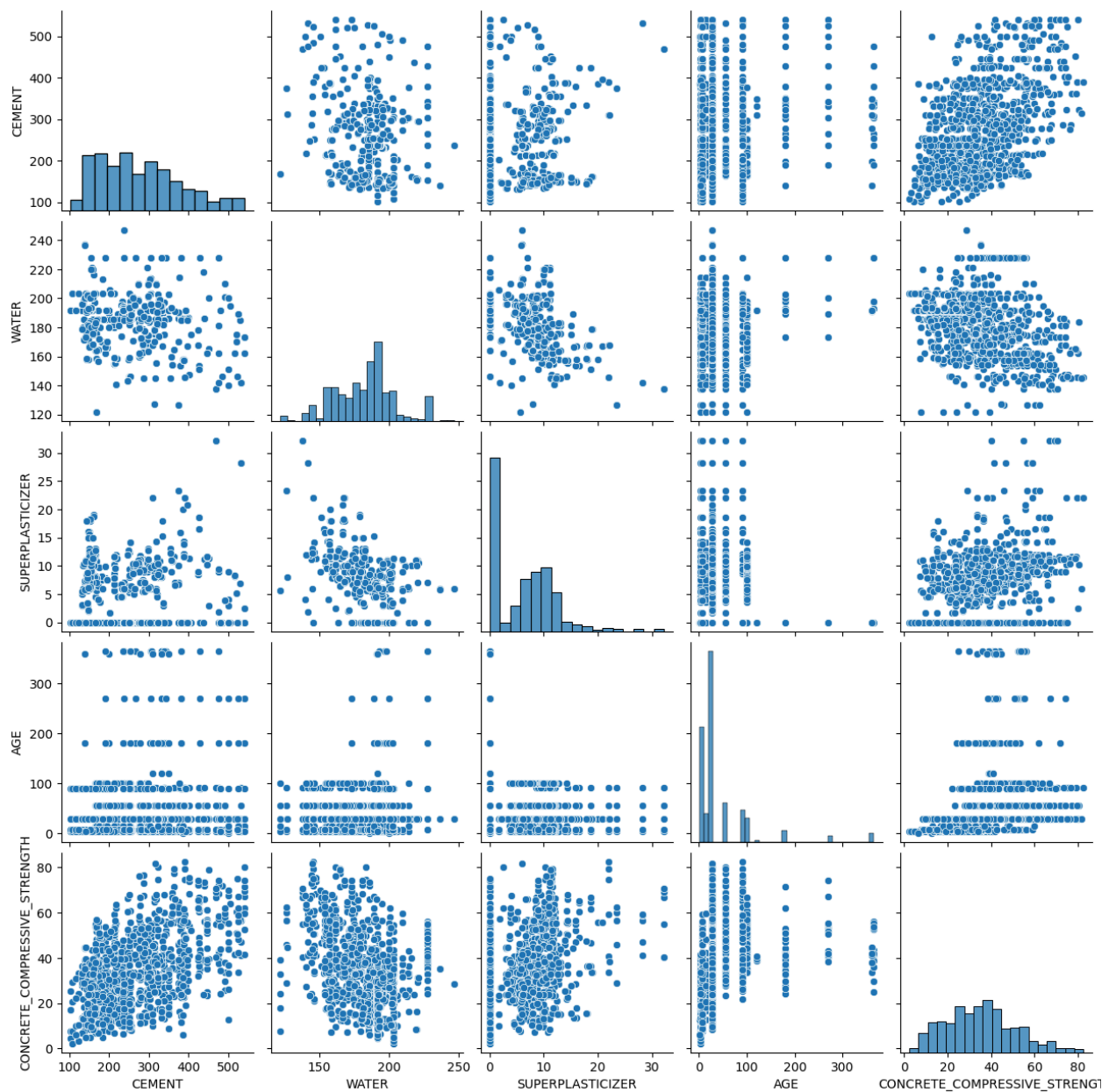


```
In [46]: # CORRELATION HEATMAP
# Shows which features are positively & negatively correlated
```

```
plt.figure(figsize=(14,10))
corr = df.corr()
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



```
In [48]: # PAIRPLOT
# Pairplot shows multiple scatter plots together
sns.pairplot(df[["CEMENT", "WATER", "SUPERPLASTICIZER", "AGE", "CONCRETE_COMPRESSIVE_STRENGTH"]])
plt.show()
```



```
In [50]: # STEP 12: Unique values
df["CONCRETE_COMPRESSIVE_STRENGTH"].unique()
```

```
Out[50]: array([79.99, 61.89, 40.27, 41.05, 44.3 , 47.03, 43.7 , 36.45, 45.85,
 39.29, 38.07, 28.02, 43.01, 42.33, 47.81, 52.91, 39.36, 56.14,
 40.56, 42.62, 41.84, 28.24, 8.06, 44.21, 52.52, 53.3 , 41.15,
 52.12, 37.43, 38.6 , 55.26, 41.72, 42.13, 53.69, 38.41, 30.08,
 37.72, 42.23, 36.25, 50.46, 39. , 53.1 , 41.54, 35.08, 15.05,
 40.76, 26.26, 32.82, 39.78, 46.93, 33.12, 49.19, 14.59, 14.64,
 41.93, 9.13, 50.95, 33.02, 54.38, 51.73, 9.87, 50.66, 48.7 ,
 55.06, 44.7 , 30.28, 40.86, 71.99, 34.4 , 28.8 , 33.4 , 36.3 ,
 29. , 37.8 , 40.2 , 28.1 , 41.3 , 25.2 , 41.1 , 35.3 , 28.3 ,
 28.6 , 24.4 , 39.3 , 40.6 , 24.1 , 46.2 , 42.8 , 49.2 , 46.8 ,
 45.7 , 55.6 , 54.9 , 34.9 , 46.9 , 54.1 , 55.9 , 49.8 , 47.1 ,
 38. , 56.1 , 59.09, 22.9 , 35.1 , 61.09, 59.8 , 60.29, 61.8 ,
 56.7 , 68.3 , 66.9 , 50.7 , 56.4 , 55.5 , 68.5 , 71.3 , 74.7 ,
 52.2 , 67.7 , 66. , 74.5 , 49.9 , 63.4 , 64.9 , 64.3 , 60.2 ,
 72.3 , 69.3 , 55.2 , 58.8 , 66.1 , 73.7 , 77.3 , 80.2 , 72.99,
 71.7 , 79.4 , 59.89, 66.6 , 65.2 , 66.7 , 62.5 , 74.19, 70.7 ,
 57.6 , 59.2 , 68.1 , 75.5 , 79.3 , 56.5 , 76.8 , 73.3 , 82.6 ,
 67.8 , 11.58, 24.45, 24.89, 29.45, 40.71, 10.38, 22.14, 22.84,
 27.66, 34.56, 12.45, 24.99, 25.72, 33.96, 37.34, 15.04, 21.06,
 26.4 , 35.34, 40.57, 12.47, 20.92, 24.9 , 34.2 , 39.61, 10.03,
 20.08, 24.48, 31.54, 9.45, 22.72, 28.47, 38.56, 40.39, 10.76,
 25.48, 21.54, 28.63, 33.54, 7.75, 17.82, 24.24, 32.85, 39.23,
 18. , 30.39, 45.71, 50.77, 53.9 , 13.18, 17.84, 40.23, 47.13,
 49.97, 13.36, 22.32, 24.54, 31.35, 19.93, 25.69, 30.23, 39.59,
 13.82, 24.92, 29.22, 38.33, 42.35, 13.54, 26.31, 31.64, 42.55,
 42.92, 13.33, 25.37, 37.4 , 44.4 , 47.74, 19.52, 38.5 , 45.08,
 47.82, 15.44, 26.77, 33.73, 42.7 , 45.84, 17.22, 29.93, 29.65,
 36.97, 43.58, 13.12, 24.43, 32.66, 36.64, 13.62, 21.6 , 27.77,
 35.57, 45.37, 7.32, 21.5 , 31.27, 43.5 , 48.67, 7.4 , 23.51,
 31.12, 39.15, 48.15, 22.5 , 34.67, 34.74, 48.97, 23.14, 41.89,
 48.28, 51.04, 55.64, 22.95, 35.23, 39.94, 48.72, 52.04, 21.02,
 33.36, 33.94, 44.14, 15.36, 28.68, 30.85, 42.03, 51.06, 21.78,
 42.29, 50.6 , 55.83, 60.95, 23.52, 42.22, 52.5 , 60.32, 66.42,
 23.8 , 38.77, 51.33, 56.85, 58.61, 21.91, 36.99, 47.4 , 51.96,
 56.74, 17.57, 40.15, 46.64, 50.08, 17.37, 33.7 , 45.94, 51.43,
 59.3 , 30.45, 47.71, 63.14, 66.82, 66.95, 27.42, 35.96, 55.51,
 61.99, 63.53, 18.02, 53.96, 56.63, 15.34, 26.05, 30.22, 37.27,
 46.23, 16.28, 25.62, 31.97, 43.06, 67.57, 57.23, 81.75, 64.02,
 78.8 , 41.37, 60.28, 56.83, 51.02, 55.55, 44.13, 39.38, 55.65,
 47.28, 44.33, 52.3 , 49.25, 29.16, 39.4 , 67.87, 58.52, 53.58,
 59. , 76.24, 69.84, 14.4 , 19.42, 20.73, 14.94, 21.29, 23.08,
 15.52, 15.82, 12.55, 8.49, 15.61, 12.18, 11.98, 16.88, 33.09,
 34.24, 31.81, 29.75, 33.01, 32.9 , 29.55, 24.66, 29.59, 24.28,
 26.2 , 46.39, 39.16, 41.2 , 33.69, 38.2 , 41.41, 37.81, 24.85,
 27.22, 44.64, 33.27, 36.56, 53.72, 48.59, 51.72, 35.85, 53.77,
 53.46, 48.99, 31.72, 39.64, 51.26, 43.39, 39.27, 37.96, 55.02,
 49.99, 53.66, 37.68, 56.06, 56.81, 50.94, 33.56, 41.16, 52.96,
 44.28, 57.03, 44.42, 53.39, 35.36, 25.02, 23.35, 52.01, 38.02,
 61.07, 55.25, 54.77, 50.24, 46.68, 22.75, 25.51, 34.77, 36.84,
 45.9 , 41.67, 56.34, 47.97, 61.46, 44.03, 55.45, 57.92, 25.61,
 33.49, 59.59, 37.92, 61.86, 62.05, 32.01, 72.1 , 65.7 , 32.11,
 40.29, 74.36, 21.97, 9.85, 15.07, 23.25, 43.73, 13.4 , 24.13,
 44.52, 62.94, 59.49, 25.12, 23.64, 35.75, 38.61, 68.75, 66.78,
 23.85, 32.07, 11.65, 19.2 , 48.85, 39.6 , 43.94, 34.57, 54.32,
```

```

15.62, 21.86, 10.22, 14.6 , 18.75, 23.4 , 25.57, 41.68, 27.74,
 8.2 , 9.62, 25.42, 15.69, 27.94, 32.63, 17.24, 19.77, 39.44,
25.75, 33.08, 24.07, 21.82, 21.07, 14.84, 32.05, 11.96, 25.45,
22.49, 25.22, 39.7 , 13.09, 38.7 , 7.51, 17.58, 21.18, 18.2 ,
17.2 , 22.63, 12.37, 25.73, 21.92, 33.04, 14.54, 26.91, 8. ,
31.9 , 10.34, 37.44, 11.48, 24.44, 17.6 , 10.73, 31.38, 13.22,
20.97, 27.04, 32.04, 35.17, 38.89, 6.47, 12.84, 18.42, 21.95,
25.08, 21.26, 25.97, 11.36, 31.25, 32.33, 9.31, 26.94, 27.63,
29.79, 34.49, 36.15, 12.54, 27.53, 32.92, 9.99, 7.84, 12.25,
11.17, 17.34, 17.54, 30.57, 14.2 , 24.5 , 15.58, 26.85, 26.06,
38.21, 30.14, 12.73, 20.87, 20.28, 34.29, 19.54, 43.38, 29.89,
6.9 , 33.19, 4.9 , 4.57, 25.46, 24.29, 33.95, 11.41, 20.59,
25.89, 29.23, 31.02, 10.39, 33.66, 27.87, 19.35, 11.39, 12.79,
39.32, 4.78, 16.11, 20.42, 6.94, 15.03, 13.57, 32.53, 15.75,
7.68, 38.8 , 33. , 17.28, 24.05, 36.59, 50.73, 13.66, 14.14,
47.78, 2.33, 16.89, 6.81, 17.96, 32.88, 22.35, 10.79, 7.72,
9.56, 6.88, 50.53, 17.17, 30.44, 9.73, 3.32, 26.32, 43.25,
6.28, 32.1 , 36.96, 54.6 , 21.48, 9.69, 8.37, 39.66, 10.09,
4.83, 10.35, 43.57, 51.86, 11.85, 27.83, 35.76, 14.31, 17.44,
31.74, 37.91, 15.87, 9.01, 33.61, 40.66, 12.05, 18.91, 25.18,
30.96, 43.89, 54.28, 36.94, 14.5 , 22.44, 12.64, 33.21, 44.09,
52.61, 59.76, 67.31, 69.66, 71.62, 74.17, 18.13, 22.53, 27.34,
29.98, 32.72, 6.27, 14.7 , 23.22, 27.92, 41.24, 14.99, 13.52,
24. , 37.42, 11.47, 21.16, 31.84, 14.8 , 21.65, 29.39, 16.26,
31.45, 37.23, 39.49, 26.74, 61.92, 47.22, 55.16, 41.64, 13.71,
19.69, 31.65, 19.11, 39.58, 48.79, 34.68, 37.33, 38.11, 33.8 ,
42.42, 48.4 , 55.94, 58.78, 67.11, 20.77, 21.75, 39.09, 24.39,
50.51, 74.99, 37.17, 33.76, 16.5 , 19.99, 36.35, 15.42, 33.42,
39.05, 27.68, 26.86, 45.3 , 30.12, 15.57, 44.61, 53.52, 57.21,
65.91, 52.82, 18.03, 37.36, 32.84, 42.64, 40.06, 41.94, 61.23,
40.87, 33.3 , 52.42, 15.09, 38.46, 37.26, 31.87, 39.45, 31.18,
23.69, 32.76, 32.4 , 36.8 , 18.28, 33.06, 31.42, 31.03, 44.39,
25.56, 36.44, 32.96, 23.84, 26.23, 17.95, 40.68, 19.01, 33.72,
8.54, 13.46, 32.24, 29.72, 49.77, 52.44, 40.93, 44.86, 13.2 ,
29.87, 56.61, 12.46, 23.79, 13.29, 39.42, 23.74, 26.14, 35.86,
28.99, 46.24, 26.92, 10.54, 25.1 , 29.07, 9.74, 39.84, 26.97,
27.23, 30.65, 33.05, 24.58, 30.88, 24.34, 23.89, 22.93, 29.41,
18.29, 28.94, 38.63, 32.25, 29.73, 52.45, 44.87, 56.62, 26.15,
15.53, 35.87, 46.25, 38.22, 39.06, 57.22, 52.83, 35.31, 43.8 ,
61.24, 33.31, 52.43, 42.14, 31.88, 39.46, 23.7 , 32.77])

```

```

In [51]: # STEP 14: # Value counts
df["CONCRETE_COMPRESSIVE_STRENGTH"].value_counts()

```

Out[51]:

count

CONCRETE_COMPRESSIVE_STRENGTH

23.52	4
31.35	4
33.40	4
41.05	4
44.52	3
43.70	3
44.28	3
40.86	3
42.13	3
39.00	3
40.93	3
25.18	3
39.30	3
18.13	3
17.54	3
32.72	3
33.80	3
41.54	3
37.43	3
35.23	3
37.27	3
12.18	3
64.90	3
46.23	3
28.63	3
9.74	2
36.45	2
28.99	2
23.74	2
13.29	2

	count
CONCRETE_COMPRESSIVE_STRENGTH	
39.42	2
23.79	2
38.60	2
10.54	2
26.92	2
31.18	2
32.40	2
36.80	2
38.46	2
42.64	2
15.09	2
40.87	2
40.06	2
53.52	2
44.61	2
15.57	2
30.12	2
45.30	2
37.36	2
18.03	2
29.07	2
13.20	2
29.87	2
49.77	2
13.46	2
8.54	2
33.72	2
19.01	2
26.23	2
23.84	2

	count
CONCRETE_COMPRESSIVE_STRENGTH	
32.96	2
36.44	2
25.56	2
31.42	2
12.46	2
44.21	2
25.10	2
27.92	2
14.99	2
13.52	2
24.00	2
37.42	2
11.47	2
37.17	2
36.30	2
36.35	2
19.99	2
33.76	2
16.50	2
15.42	2
26.86	2
27.68	2
33.42	2
65.91	2
14.20	2
43.38	2
26.06	2
17.24	2
41.68	2
19.77	2

	count
CONCRETE_COMPRESSIVE_STRENGTH	
38.70	2
36.94	2
22.44	2
43.57	2
37.91	2
17.96	2
24.40	2
24.10	2
54.90	2
29.55	2
29.59	2
24.28	2
33.69	2
46.68	2
37.81	2
21.86	2
37.92	2
39.70	2
55.55	2
51.02	2
52.20	2
19.42	2
39.38	2
15.52	2
20.73	2
40.15	2
33.70	2
47.71	2
15.34	2
41.37	2

	count
CONCRETE_COMPRESSIVE_STRENGTH	
31.97	2
21.91	2
51.04	2
45.08	2
33.73	2
43.58	2
45.37	2
35.34	2
44.30	2
56.14	2
52.91	2
45.85	1
39.29	1
38.07	1
28.02	1
43.01	1
24.58	1
35.87	1
15.53	1
42.33	1
32.77	1
33.05	1
30.65	1
27.23	1
26.97	1
39.84	1
46.24	1
53.69	1
41.72	1
55.26	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
40.27	1
52.12	1
41.15	1
53.30	1
52.52	1
8.06	1
28.24	1
22.14	1
10.38	1
40.71	1
29.45	1
24.89	1
24.45	1
11.58	1
67.80	1
82.60	1
73.30	1
76.80	1
56.50	1
79.30	1
75.50	1
68.10	1
57.22	1
34.20	1
24.90	1
20.92	1
12.47	1
40.57	1
26.40	1
21.06	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
15.04	1
37.34	1
33.96	1
25.72	1
24.99	1
12.45	1
34.56	1
27.66	1
22.84	1
17.82	1
7.75	1
33.54	1
21.54	1
25.48	1
10.76	1
40.39	1
38.56	1
28.47	1
22.72	1
9.45	1
31.54	1
24.48	1
20.08	1
10.03	1
39.61	1
24.54	1
22.32	1
13.36	1
49.97	1
47.13	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
40.23	1
17.84	1
13.18	1
53.90	1
50.77	1
45.71	1
30.39	1
18.00	1
39.23	1
32.85	1
24.24	1
25.37	1
13.33	1
42.92	1
42.55	1
31.64	1
26.31	1
13.54	1
42.35	1
38.33	1
29.22	1
24.92	1
13.82	1
39.59	1
30.23	1
25.69	1
19.93	1
24.43	1
13.12	1
36.97	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
29.65	1
29.93	1
17.22	1
45.84	1
42.70	1
26.77	1
15.44	1
47.82	1
38.50	1
19.52	1
47.74	1
44.40	1
37.40	1
48.15	1
39.15	1
31.12	1
23.51	1
7.40	1
48.67	1
43.50	1
31.27	1
21.50	1
7.32	1
35.57	1
27.77	1
21.60	1
13.62	1
36.64	1
32.66	1
44.14	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
33.94	1
33.36	1
21.02	1
52.04	1
48.72	1
39.94	1
22.95	1
55.64	1
48.28	1
41.89	1
23.14	1
48.97	1
34.74	1
34.67	1
22.50	1
38.77	1
23.80	1
66.42	1
60.32	1
52.50	1
42.22	1
60.95	1
55.83	1
50.60	1
42.29	1
21.78	1
51.06	1
42.03	1
30.85	1
28.68	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
15.36	1
65.20	1
66.60	1
59.89	1
79.40	1
71.70	1
72.99	1
80.20	1
77.30	1
73.70	1
66.10	1
51.96	1
47.40	1
36.99	1
58.61	1
56.85	1
51.33	1
63.14	1
30.45	1
59.30	1
51.43	1
45.94	1
17.37	1
50.08	1
46.64	1
17.57	1
56.74	1
59.20	1
57.60	1
70.70	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
74.19	1
62.50	1
66.70	1
67.57	1
43.06	1
25.62	1
16.28	1
30.22	1
26.05	1
56.63	1
53.96	1
18.02	1
63.53	1
61.99	1
55.51	1
35.96	1
27.42	1
66.95	1
66.82	1
58.52	1
67.87	1
39.40	1
29.16	1
49.25	1
52.30	1
44.33	1
47.28	1
55.65	1
44.13	1
56.83	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
60.28	1
78.80	1
64.02	1
81.75	1
57.23	1
63.40	1
49.90	1
74.50	1
66.00	1
67.70	1
74.70	1
71.30	1
68.50	1
23.08	1
21.29	1
14.94	1
14.40	1
69.84	1
76.24	1
59.00	1
53.58	1
29.75	1
31.81	1
34.24	1
33.09	1
16.88	1
11.98	1
15.61	1
8.49	1
12.55	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
15.82	1
58.80	1
55.20	1
69.30	1
72.30	1
60.20	1
64.30	1
48.59	1
53.72	1
36.56	1
33.27	1
44.64	1
27.22	1
24.85	1
41.41	1
38.20	1
41.20	1
39.16	1
46.39	1
26.20	1
24.66	1
32.90	1
33.01	1
56.06	1
37.68	1
53.66	1
49.99	1
55.02	1
37.96	1
39.27	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
43.39	1
51.26	1
39.64	1
31.72	1
48.99	1
53.46	1
53.77	1
35.85	1
51.72	1
54.77	1
55.25	1
61.07	1
38.02	1
52.01	1
23.35	1
25.02	1
35.36	1
53.39	1
44.42	1
57.03	1
52.96	1
41.16	1
33.56	1
50.94	1
56.81	1
59.59	1
33.49	1
25.61	1
57.92	1
55.45	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
44.03	1
61.46	1
47.97	1
56.34	1
41.67	1
45.90	1
36.84	1
34.77	1
25.51	1
22.75	1
50.24	1
62.94	1
24.13	1
13.40	1
43.73	1
23.25	1
15.07	1
9.85	1
21.97	1
74.36	1
40.29	1
32.11	1
65.70	1
72.10	1
32.01	1
62.05	1
61.86	1
54.32	1
34.57	1
43.94	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
39.60	1
48.85	1
19.20	1
11.65	1
32.07	1
23.85	1
66.78	1
68.75	1
38.61	1
35.75	1
23.64	1
25.12	1
59.49	1
33.08	1
25.75	1
39.44	1
32.63	1
27.94	1
15.69	1
25.42	1
9.62	1
8.20	1
27.74	1
25.57	1
23.40	1
18.75	1
14.60	1
10.22	1
15.62	1
22.63	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
17.20	1
18.20	1
21.18	1
17.58	1
7.51	1
13.09	1
25.22	1
22.49	1
25.45	1
11.96	1
32.05	1
14.84	1
21.07	1
21.82	1
24.07	1
13.22	1
31.38	1
10.73	1
17.60	1
24.44	1
11.48	1
37.44	1
10.34	1
31.90	1
8.00	1
26.91	1
14.54	1
33.04	1
21.92	1
25.73	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
12.37	1
9.31	1
32.33	1
31.25	1
11.36	1
25.97	1
21.26	1
25.08	1
21.95	1
18.42	1
12.84	1
6.47	1
38.89	1
35.17	1
32.04	1
27.04	1
20.97	1
61.80	1
60.29	1
59.80	1
61.09	1
35.10	1
22.90	1
59.09	1
56.10	1
38.00	1
47.10	1
12.54	1
36.15	1
34.49	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
29.79	1
27.63	1
26.94	1
15.58	1
24.50	1
30.57	1
17.34	1
11.17	1
12.25	1
7.84	1
9.99	1
32.92	1
27.53	1
55.50	1
56.40	1
50.70	1
66.90	1
68.30	1
56.70	1
33.95	1
24.29	1
25.46	1
4.57	1
4.90	1
33.19	1
6.90	1
29.89	1
19.54	1
34.29	1
20.28	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
20.87	1
12.73	1
30.14	1
38.21	1
26.85	1
6.94	1
20.42	1
16.11	1
4.78	1
39.32	1
12.79	1
11.39	1
19.35	1
27.87	1
33.66	1
10.39	1
31.02	1
29.23	1
25.89	1
20.59	1
11.41	1
16.89	1
2.33	1
47.78	1
14.14	1
13.66	1
50.73	1
36.59	1
24.05	1
17.28	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
33.00	1
38.80	1
7.68	1
15.75	1
32.53	1
13.57	1
15.03	1
32.10	1
6.28	1
43.25	1
26.32	1
3.32	1
9.73	1
30.44	1
17.17	1
50.53	1
6.88	1
9.56	1
7.72	1
10.79	1
22.35	1
32.88	1
6.81	1
31.74	1
17.44	1
14.31	1
35.76	1
27.83	1
11.85	1
51.86	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
10.35	1
4.83	1
10.09	1
39.66	1
8.37	1
9.69	1
21.48	1
54.60	1
36.96	1
49.20	1
42.80	1
46.20	1
40.60	1
33.21	1
12.64	1
14.50	1
54.28	1
43.89	1
30.96	1
18.91	1
12.05	1
40.66	1
33.61	1
9.01	1
15.87	1
22.53	1
74.17	1
71.62	1
69.66	1
67.31	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
59.76	1
52.61	1
44.09	1
49.80	1
55.90	1
54.10	1
46.90	1
34.90	1
55.60	1
45.70	1
46.80	1
26.74	1
39.49	1
37.23	1
31.45	1
16.26	1
29.39	1
21.65	1
14.80	1
31.84	1
21.16	1
41.24	1
23.22	1
14.70	1
6.27	1
29.98	1
27.34	1
71.99	1
48.40	1
42.42	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
38.11	1
37.33	1
34.68	1
48.79	1
39.58	1
19.11	1
31.65	1
19.69	1
13.71	1
41.64	1
55.16	1
47.22	1
61.92	1
20.77	1
67.11	1
58.78	1
55.94	1
28.60	1
28.30	1
35.30	1
41.10	1
25.20	1
41.30	1
28.10	1
40.20	1
37.80	1
29.00	1
28.80	1
34.40	1
50.95	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
9.13	1
41.93	1
14.64	1
14.59	1
49.19	1
33.12	1
32.84	1
52.82	1
57.21	1
39.05	1
74.99	1
50.51	1
24.39	1
39.09	1
21.75	1
39.45	1
31.87	1
37.26	1
52.42	1
33.30	1
61.23	1
41.94	1
30.28	1
44.70	1
55.06	1
48.70	1
50.66	1
9.87	1
51.73	1
54.38	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
33.02	1
53.10	1
50.46	1
36.25	1
42.23	1
37.72	1
30.08	1
38.41	1
32.24	1
40.68	1
17.95	1
44.39	1
31.03	1
33.06	1
18.28	1
32.76	1
23.69	1
41.84	1
42.62	1
40.56	1
35.86	1
26.14	1
56.61	1
44.86	1
52.44	1
29.72	1
46.93	1
39.78	1
32.82	1
26.26	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
40.76	1
15.05	1
35.08	1
79.99	1
23.70	1
39.46	1
31.88	1
42.14	1
52.43	1
33.31	1
61.24	1
43.80	1
35.31	1
52.83	1
22.93	1
23.89	1
24.34	1
30.88	1
61.89	1
39.06	1
38.22	1
46.25	1
26.15	1
56.62	1
44.87	1
52.45	1
29.73	1
32.25	1
38.63	1
28.94	1

	count
CONCRETE_COMPRESSIVE_STRENGTH	
18.29	1
29.41	1
39.36	1
47.81	1
47.03	1

dtype: int64

```
In [52]: # Unique values of every column
df.nunique()
```

```
Out[52]:
```

	0
CEMENT	278
BLAST_FURNACE_SLAG	185
FLY_ASH	156
WATER	195
SUPERPLASTICIZER	111
COARSE_AGGREGATE	284
FINE_AGGREGATE	302
AGE	14
CONCRETE_COMPRESSIVE_STRENGTH	845

dtype: int64

```
In [53]: # Value counts for CEMENT
df["CEMENT"].value_counts()
```

Out[53]:

count	
CEMENT	
251.4	15
446.0	14
310.0	14
250.0	13
475.0	13
331.0	13
349.0	12
387.0	12
236.0	11
165.0	11
339.0	10
425.0	10
213.7	10
213.5	10
500.0	10
380.0	10
190.3	10
194.7	10
540.0	9
212.0	9
359.0	8
350.0	8
424.0	8
491.0	8
153.0	8
277.0	8
313.3	8
480.0	7
238.1	7
296.0	7

	count
CEMENT	
375.0	7
525.0	7
393.0	7
200.0	7
203.5	6
362.6	6
313.0	6
289.0	6
385.0	6
284.0	6
167.0	6
139.6	6
237.5	6
374.0	6
427.5	6
304.0	6
266.0	6
198.6	6
332.5	6
190.0	5
218.2	5
214.9	5
231.8	5
181.4	5
218.9	5
376.0	5
251.8	5
295.7	5
277.2	5
297.2	5

	count
CEMENT	
275.1	5
246.8	5
252.3	5
249.1	5
168.9	5
182.0	5
277.1	5
290.4	5
307.0	5
255.0	5
167.4	5
178.0	5
173.5	5
172.4	5
173.8	5
233.8	5
212.1	5
190.7	5
168.0	5
213.8	5
230.0	5
166.1	5
229.7	5
212.5	5
397.0	5
333.0	5
212.6	5
439.0	5
286.3	5
379.5	5

	count
CEMENT	
149.0	5
222.4	5
337.9	5
155.0	5
389.9	5
318.8	5
531.3	5
322.0	5
382.0	5
288.0	5
401.8	5
388.6	5
469.0	5
157.0	5
133.0	5
342.0	4
254.0	4
173.0	4
250.2	4
202.0	4
141.3	4
183.9	4
102.0	4
144.0	4
252.0	4
298.0	4
108.3	4
305.3	4
122.6	4
166.8	4

	count
CEMENT	
162.0	4
192.0	4
116.0	4
323.7	4
154.8	4
272.8	3
275.0	3
300.0	3
314.0	3
302.0	3
238.0	3
281.0	3
160.0	3
145.0	3
159.0	3
148.0	3
382.5	2
210.7	2
381.4	2
228.0	2
316.1	2
135.7	2
325.0	2
295.8	2
186.2	2
255.5	2
239.6	2
158.8	2
238.2	2
355.0	2

	count
CEMENT	
220.8	2
181.9	2
356.0	2
520.0	2
516.0	2
193.5	2
170.3	2
225.0	2
295.0	2
273.0	2
164.0	2
326.0	2
154.0	2
152.0	2
156.0	2
366.0	2
146.0	2
132.0	2
166.0	2
140.0	2
321.0	2
150.0	2
312.7	2
298.1	2
148.1	2
135.0	2
153.1	2
136.0	2
339.2	2
290.2	2

	count
CEMENT	
252.5	2
400.0	2
143.0	2
151.0	2
522.0	2
405.0	1
436.0	1
505.0	1
334.0	1
234.0	1
315.0	1
450.1	1
500.1	1
325.6	1
451.0	1
528.0	1
285.0	1
485.0	1
164.6	1
252.1	1
298.2	1
264.5	1
321.4	1
313.8	1
139.7	1
288.4	1
287.3	1
317.9	1
355.9	1
164.2	1

	count
CEMENT	
160.2	1
155.6	1
322.5	1
305.0	1
261.0	1
237.0	1
265.0	1
280.0	1
287.0	1
318.0	1
276.0	1
142.0	1
297.0	1
147.0	1
260.0	1
336.0	1
236.9	1
145.9	1
262.0	1
184.0	1
151.6	1
139.9	1
149.5	1
299.8	1
326.5	1
152.7	1
261.9	1
158.4	1
150.7	1
272.6	1

	count
CEMENT	
259.9	1
312.9	1
336.5	1
144.8	1
133.1	1
158.0	1
330.5	1
143.7	1
266.2	1
134.7	1
145.7	1
143.8	1
155.2	1
147.8	1
145.4	1
136.4	1
255.3	1
153.6	1
146.5	1
151.8	1
309.9	1
143.6	1
303.6	1
374.3	1
158.6	1
152.6	1
304.8	1
150.9	1
141.9	1
297.8	1

	count
CEMENT	
321.3	1
279.8	1
159.8	1
276.4	1
322.2	1
148.5	1
159.1	1
260.9	1

dtype: int64

```
In [54]: # STEP 13: RE CHECK
df.head()
```

```
Out[54]:
```

	CEMENT	BLAST_FURNACE_SLAG	FLY_ASH	WATER	SUPERPLASTICIZER	COAR
0	540.0	0.0	0.0	162.0	2.5	
1	540.0	0.0	0.0	162.0	2.5	
2	332.5	142.5	0.0	228.0	0.0	
3	332.5	142.5	0.0	228.0	0.0	
4	198.6	132.4	0.0	192.0	0.0	

STEP-2: DEEP LEARNING (ANN)

```
In [55]: # STEP 1: LOAD REQUIRED LIB
from sklearn.model_selection import train_test_split # for train-test split
from sklearn.metrics import mean_absolute_error, mean_squared_error # for c
from sklearn.ensemble import ExtraTreesRegressor # for feature importance
```

```
In [56]: # STEP 2 : DEFINE X AND Y
X = df.iloc[:, :-1] # all columns except last
y = df.iloc[:, -1] # last column is STRENGTH
```

```
In [58]: # STEP 3 : CORRELATION MATRIX
df.corr() # numerical correlation between columns
```

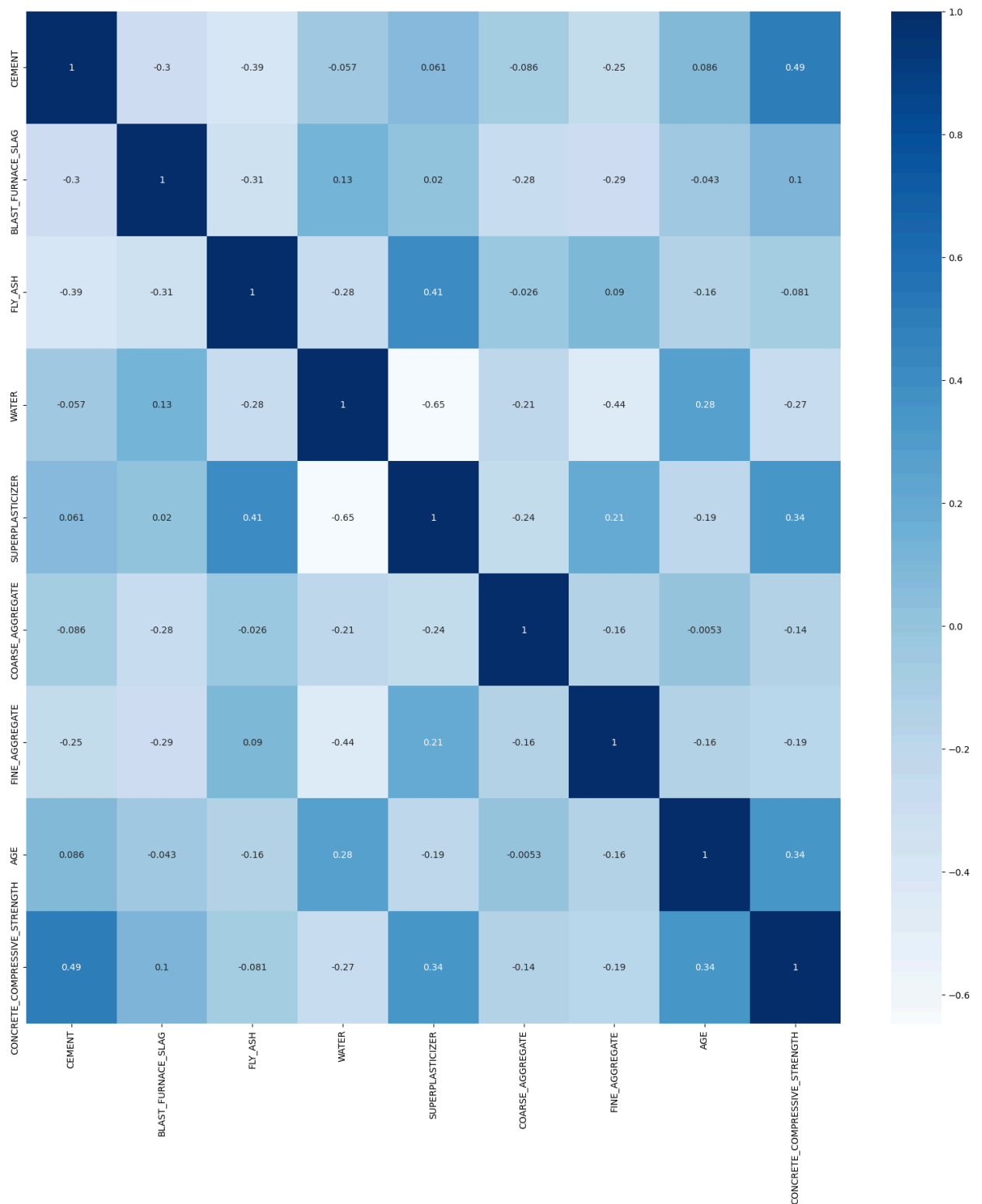
Out[58]:

	CEMENT	BLAST_FURNACE_SLAG	FLY_ASH
CEMENT	1.000000	-0.303324	-0.385610
BLAST_FURNACE_SLAG	-0.303324	1.000000	-0.312352
FLY_ASH	-0.385610	-0.312352	1.000000
WATER	-0.056625	0.130262	-0.283314
SUPERPLASTICIZER	0.060906	0.019800	0.414213
COARSE_AGGREGATE	-0.086205	-0.277559	-0.026468
FINE_AGGREGATE	-0.245375	-0.289685	0.090262
AGE	0.086348	-0.042759	-0.158940
CONCRETE_COMPRESSIVE_STRENGTH	0.488283	0.103374	-0.080648

In [60]: *# STEP 4: HEATMAP OF CORRELATION MATRIX*

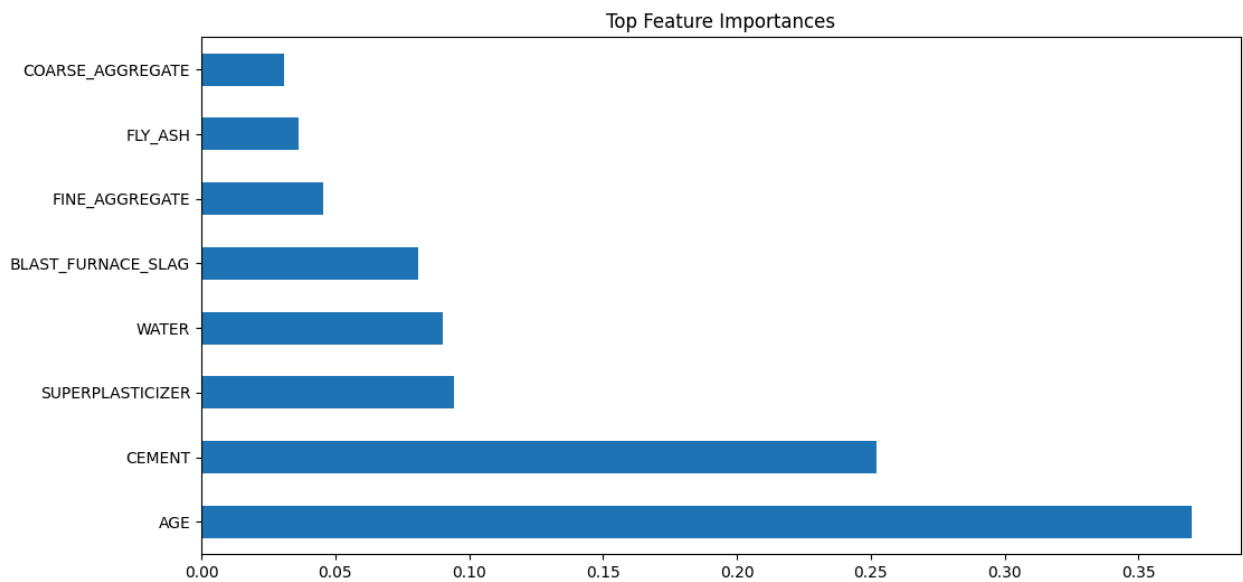
```
corrmat = df.corr() # calculate correlation
top_corr_features = corrmat.index # get column names

plt.figure(figsize=(20,20))
sns.heatmap(df[top_corr_features].corr(), annot=True, cmap="Blues")
plt.show()
```



```
In [79]: # STEP 5 : FEATURE IMPORTANCE USING EXTRATREES
model = ExtraTreesRegressor()
model.fit(X, y)
plt.figure(figsize=(12,6))
feat_importances = pd.Series(model.feature_importances_, index=X.columns)
feat_importances.nlargest(10).plot(kind='barh') # show top important featur
```

```
plt.title("Top Feature Importances")
plt.show()
```



```
In [78]: # STEP 6 : DISTRIBUTION OF TARGET VARIABLE
plt.figure(figsize=(12,6))
sns.distplot(y)
plt.title("Distribution of Strength")
plt.show()
```

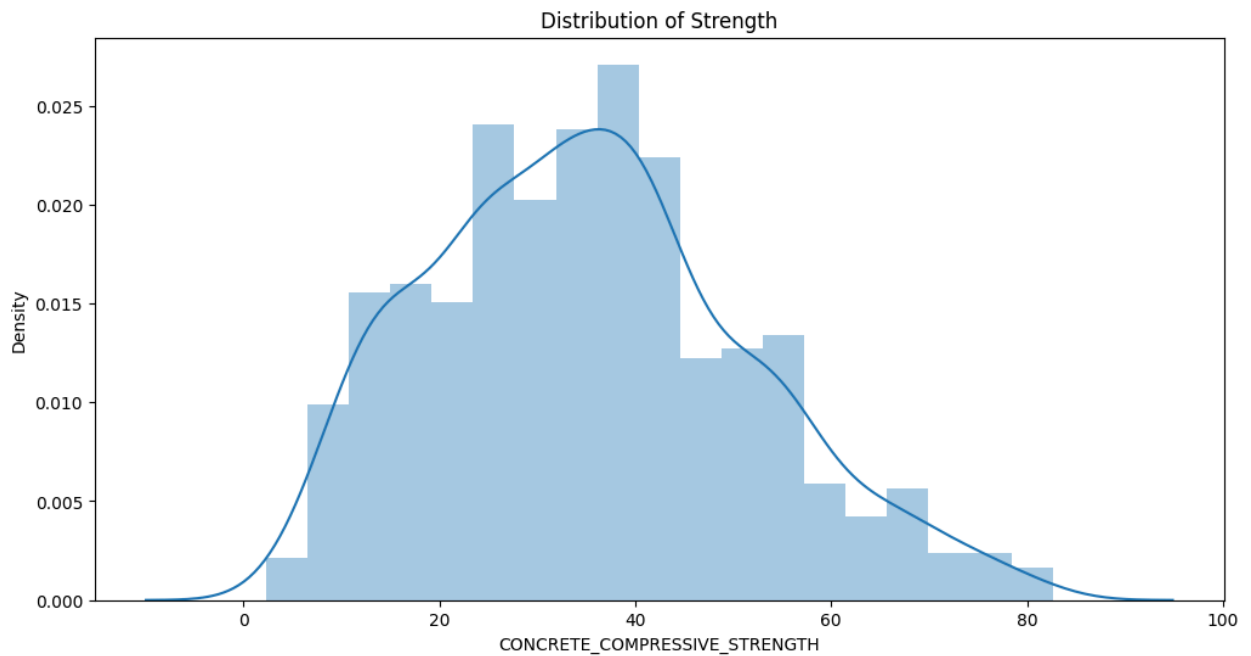
/tmp/ipython-input-2636715438.py:3: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(y)
```



```
In [64]: # STEP 7 : TRAIN-TEST SPLIT ( TRAIN= 70% AND TEST= 30%)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
```

```
In [66]: # STEP 8: BUILD ANN MODEL
from keras.models import Sequential
from keras.layers import Dense, Dropout
from keras.layers import LeakyReLU, PReLU, ELU
```

```
In [67]: NN_model = Sequential()    # create empty neural network
```

```
In [69]: # Input Layer
NN_model.add(Dense(128, kernel_initializer='normal', input_dim=X_train.shape[1], activation='relu'))
```

```
In [70]: # Hidden Layers
NN_model.add(Dense(256, kernel_initializer='normal', activation='relu')) # first hidden layer
NN_model.add(Dense(256, kernel_initializer='normal', activation='relu')) # second hidden layer
NN_model.add(Dense(256, kernel_initializer='normal', activation='relu')) # third hidden layer
```

```
In [71]: # Output Layer
NN_model.add(Dense(1, kernel_initializer='normal', activation='linear'))
```

```
In [72]: # STEP 9: COMPILE THE MODEL
NN_model.compile(
    loss='mean_absolute_error',
    optimizer='adam',
    metrics=['mean_absolute_error'])
```

```
In [73]: NN_model.summary()    # show full model structure
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 128)	1,152
dense_1 (Dense)	(None, 128)	16,512
dense_2 (Dense)	(None, 256)	33,024
dense_3 (Dense)	(None, 256)	65,792
dense_4 (Dense)	(None, 256)	65,792
dense_5 (Dense)	(None, 1)	257

Total params: 182,529 (713.00 KB)

Trainable params: 182,529 (713.00 KB)

Non-trainable params: 0 (0.00 B)

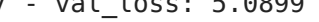
```
In [74]: # STEP 10: TRAIN THE MODEL
model_history = NN_model.fit(X_train, y_train, validation_split=0.33, batch_size
```

Epoch 1/100
48/48  3s 14ms/step - loss: 17.6513 - mean_absolute_error: 17.6513 - val_loss: 13.2158 - val_mean_absolute_error: 13.2158
Epoch 2/100
48/48  0s 9ms/step - loss: 12.4250 - mean_absolute_error: 12.4250 - val_loss: 10.2999 - val_mean_absolute_error: 10.2999
Epoch 3/100
48/48  0s 10ms/step - loss: 11.4765 - mean_absolute_error: 11.4765 - val_loss: 10.8955 - val_mean_absolute_error: 10.8955
Epoch 4/100
48/48  0s 9ms/step - loss: 10.8643 - mean_absolute_error: 10.8643 - val_loss: 9.2130 - val_mean_absolute_error: 9.2130
Epoch 5/100
48/48  0s 5ms/step - loss: 9.0057 - mean_absolute_error: 9.0057 - val_loss: 8.1850 - val_mean_absolute_error: 8.1850
Epoch 6/100
48/48  0s 6ms/step - loss: 8.8538 - mean_absolute_error: 8.8538 - val_loss: 7.6035 - val_mean_absolute_error: 7.6035
Epoch 7/100
48/48  0s 5ms/step - loss: 7.6418 - mean_absolute_error: 7.6418 - val_loss: 11.4432 - val_mean_absolute_error: 11.4432
Epoch 8/100
48/48  0s 5ms/step - loss: 8.6664 - mean_absolute_error: 8.6664 - val_loss: 6.1607 - val_mean_absolute_error: 6.1607
Epoch 9/100
48/48  0s 5ms/step - loss: 6.8777 - mean_absolute_error: 6.8777 - val_loss: 6.4061 - val_mean_absolute_error: 6.4061
Epoch 10/100
48/48  0s 5ms/step - loss: 7.0682 - mean_absolute_error: 7.0682 - val_loss: 6.4829 - val_mean_absolute_error: 6.4829
Epoch 11/100
48/48  0s 5ms/step - loss: 7.8358 - mean_absolute_error: 7.8358 - val_loss: 7.4910 - val_mean_absolute_error: 7.4910
Epoch 12/100
48/48  0s 5ms/step - loss: 7.1723 - mean_absolute_error: 7.1723 - val_loss: 6.6755 - val_mean_absolute_error: 6.6755
Epoch 13/100
48/48  0s 5ms/step - loss: 6.3506 - mean_absolute_error: 6.3506 - val_loss: 9.9337 - val_mean_absolute_error: 9.9337
Epoch 14/100
48/48  0s 5ms/step - loss: 7.5418 - mean_absolute_error: 7.5418 - val_loss: 7.6332 - val_mean_absolute_error: 7.6332
Epoch 15/100
48/48  0s 5ms/step - loss: 7.1484 - mean_absolute_error: 7.1484 - val_loss: 5.2200 - val_mean_absolute_error: 5.2200
Epoch 16/100
48/48  0s 5ms/step - loss: 6.2703 - mean_absolute_error: 6.2703 - val_loss: 5.5493 - val_mean_absolute_error: 5.5493
Epoch 17/100
48/48  0s 5ms/step - loss: 6.7858 - mean_absolute_error: 6.7858 - val_loss: 5.4322 - val_mean_absolute_error: 5.4322
Epoch 18/100
48/48  0s 5ms/step - loss: 6.2370 - mean_absolute_error: 6.2370 - val_loss: 5.9074 - val_mean_absolute_error: 5.9074

Epoch 19/100
48/48  0s 5ms/step - loss: 6.4608 - mean_absolute_error: 6.4608 - val_loss: 7.7261 - val_mean_absolute_error: 7.7261
Epoch 20/100
48/48  0s 5ms/step - loss: 7.2300 - mean_absolute_error: 7.2300 - val_loss: 5.0930 - val_mean_absolute_error: 5.0930
Epoch 21/100
48/48  0s 6ms/step - loss: 6.3183 - mean_absolute_error: 6.3183 - val_loss: 7.5777 - val_mean_absolute_error: 7.5777
Epoch 22/100
48/48  0s 5ms/step - loss: 6.6042 - mean_absolute_error: 6.6042 - val_loss: 5.2160 - val_mean_absolute_error: 5.2160
Epoch 23/100
48/48  0s 5ms/step - loss: 6.7438 - mean_absolute_error: 6.7438 - val_loss: 8.2912 - val_mean_absolute_error: 8.2912
Epoch 24/100
48/48  0s 6ms/step - loss: 7.1208 - mean_absolute_error: 7.1208 - val_loss: 5.1822 - val_mean_absolute_error: 5.1822
Epoch 25/100
48/48  0s 5ms/step - loss: 6.7588 - mean_absolute_error: 6.7588 - val_loss: 5.7098 - val_mean_absolute_error: 5.7098
Epoch 26/100
48/48  0s 5ms/step - loss: 6.2578 - mean_absolute_error: 6.2578 - val_loss: 4.8276 - val_mean_absolute_error: 4.8276
Epoch 27/100
48/48  0s 5ms/step - loss: 6.0101 - mean_absolute_error: 6.0101 - val_loss: 6.6776 - val_mean_absolute_error: 6.6776
Epoch 28/100
48/48  0s 5ms/step - loss: 6.0201 - mean_absolute_error: 6.0201 - val_loss: 4.9293 - val_mean_absolute_error: 4.9293
Epoch 29/100
48/48  0s 5ms/step - loss: 6.4884 - mean_absolute_error: 6.4884 - val_loss: 7.3496 - val_mean_absolute_error: 7.3496
Epoch 30/100
48/48  0s 5ms/step - loss: 5.8118 - mean_absolute_error: 5.8118 - val_loss: 4.9193 - val_mean_absolute_error: 4.9193
Epoch 31/100
48/48  0s 5ms/step - loss: 5.4505 - mean_absolute_error: 5.4505 - val_loss: 7.2094 - val_mean_absolute_error: 7.2094
Epoch 32/100
48/48  0s 6ms/step - loss: 6.3537 - mean_absolute_error: 6.3537 - val_loss: 6.6055 - val_mean_absolute_error: 6.6055
Epoch 33/100
48/48  0s 6ms/step - loss: 6.3752 - mean_absolute_error: 6.3752 - val_loss: 4.9415 - val_mean_absolute_error: 4.9415
Epoch 34/100
48/48  0s 5ms/step - loss: 5.0564 - mean_absolute_error: 5.0564 - val_loss: 4.7921 - val_mean_absolute_error: 4.7921
Epoch 35/100
48/48  0s 5ms/step - loss: 5.4859 - mean_absolute_error: 5.4859 - val_loss: 4.9703 - val_mean_absolute_error: 4.9703
Epoch 36/100
48/48  0s 5ms/step - loss: 6.0671 - mean_absolute_error: 6.0671 - val_loss: 5.9919 - val_mean_absolute_error: 5.9919

Epoch 37/100
48/48 ————— 0s 5ms/step - loss: 5.7493 - mean_absolute_error: 5.7493 - val_loss: 5.9961 - val_mean_absolute_error: 5.9961
Epoch 38/100
48/48 ————— 0s 5ms/step - loss: 5.7890 - mean_absolute_error: 5.7890 - val_loss: 6.2065 - val_mean_absolute_error: 6.2065
Epoch 39/100
48/48 ————— 0s 7ms/step - loss: 5.7907 - mean_absolute_error: 5.7907 - val_loss: 6.4170 - val_mean_absolute_error: 6.4170
Epoch 40/100
48/48 ————— 0s 8ms/step - loss: 6.6013 - mean_absolute_error: 6.6013 - val_loss: 5.6913 - val_mean_absolute_error: 5.6913
Epoch 41/100
48/48 ————— 0s 8ms/step - loss: 5.1540 - mean_absolute_error: 5.1540 - val_loss: 7.2789 - val_mean_absolute_error: 7.2789
Epoch 42/100
48/48 ————— 1s 10ms/step - loss: 5.9838 - mean_absolute_error: 5.9838 - val_loss: 7.4139 - val_mean_absolute_error: 7.4139
Epoch 43/100
48/48 ————— 1s 10ms/step - loss: 5.7162 - mean_absolute_error: 5.7162 - val_loss: 5.5987 - val_mean_absolute_error: 5.5987
Epoch 44/100
48/48 ————— 0s 7ms/step - loss: 6.4164 - mean_absolute_error: 6.4164 - val_loss: 5.7862 - val_mean_absolute_error: 5.7862
Epoch 45/100
48/48 ————— 0s 5ms/step - loss: 5.6886 - mean_absolute_error: 5.6886 - val_loss: 4.3564 - val_mean_absolute_error: 4.3564
Epoch 46/100
48/48 ————— 0s 5ms/step - loss: 5.0361 - mean_absolute_error: 5.0361 - val_loss: 5.0196 - val_mean_absolute_error: 5.0196
Epoch 47/100
48/48 ————— 0s 6ms/step - loss: 5.2175 - mean_absolute_error: 5.2175 - val_loss: 5.1139 - val_mean_absolute_error: 5.1139
Epoch 48/100
48/48 ————— 0s 5ms/step - loss: 6.0628 - mean_absolute_error: 6.0628 - val_loss: 6.1287 - val_mean_absolute_error: 6.1287
Epoch 49/100
48/48 ————— 0s 5ms/step - loss: 6.1881 - mean_absolute_error: 6.1881 - val_loss: 4.5378 - val_mean_absolute_error: 4.5378
Epoch 50/100
48/48 ————— 0s 6ms/step - loss: 5.2492 - mean_absolute_error: 5.2492 - val_loss: 5.3398 - val_mean_absolute_error: 5.3398
Epoch 51/100
48/48 ————— 0s 5ms/step - loss: 6.3401 - mean_absolute_error: 6.3401 - val_loss: 5.6529 - val_mean_absolute_error: 5.6529
Epoch 52/100
48/48 ————— 0s 5ms/step - loss: 5.8753 - mean_absolute_error: 5.8753 - val_loss: 4.8036 - val_mean_absolute_error: 4.8036
Epoch 53/100
48/48 ————— 0s 5ms/step - loss: 5.0084 - mean_absolute_error: 5.0084 - val_loss: 5.3452 - val_mean_absolute_error: 5.3452
Epoch 54/100
48/48 ————— 0s 6ms/step - loss: 4.9209 - mean_absolute_error: 4.9209 - val_loss: 4.4824 - val_mean_absolute_error: 4.4824

Epoch 55/100
48/48 ————— 0s 5ms/step - loss: 5.1443 - mean_absolute_error: 5.1443 - val_loss: 4.6523 - val_mean_absolute_error: 4.6523
Epoch 56/100
48/48 ————— 0s 5ms/step - loss: 4.8675 - mean_absolute_error: 4.8675 - val_loss: 5.1898 - val_mean_absolute_error: 5.1898
Epoch 57/100
48/48 ————— 0s 5ms/step - loss: 4.7318 - mean_absolute_error: 4.7318 - val_loss: 4.8321 - val_mean_absolute_error: 4.8321
Epoch 58/100
48/48 ————— 0s 6ms/step - loss: 4.5948 - mean_absolute_error: 4.5948 - val_loss: 5.2893 - val_mean_absolute_error: 5.2893
Epoch 59/100
48/48 ————— 0s 5ms/step - loss: 4.9099 - mean_absolute_error: 4.9099 - val_loss: 4.9523 - val_mean_absolute_error: 4.9523
Epoch 60/100
48/48 ————— 0s 5ms/step - loss: 4.6512 - mean_absolute_error: 4.6512 - val_loss: 5.7722 - val_mean_absolute_error: 5.7722
Epoch 61/100
48/48 ————— 0s 5ms/step - loss: 5.3869 - mean_absolute_error: 5.3869 - val_loss: 4.6074 - val_mean_absolute_error: 4.6074
Epoch 62/100
48/48 ————— 0s 5ms/step - loss: 5.1958 - mean_absolute_error: 5.1958 - val_loss: 6.0628 - val_mean_absolute_error: 6.0628
Epoch 63/100
48/48 ————— 0s 5ms/step - loss: 5.8172 - mean_absolute_error: 5.8172 - val_loss: 5.8439 - val_mean_absolute_error: 5.8439
Epoch 64/100
48/48 ————— 0s 5ms/step - loss: 5.2707 - mean_absolute_error: 5.2707 - val_loss: 4.5399 - val_mean_absolute_error: 4.5399
Epoch 65/100
48/48 ————— 0s 6ms/step - loss: 5.1775 - mean_absolute_error: 5.1775 - val_loss: 8.3527 - val_mean_absolute_error: 8.3527
Epoch 66/100
48/48 ————— 0s 5ms/step - loss: 6.3752 - mean_absolute_error: 6.3752 - val_loss: 4.7264 - val_mean_absolute_error: 4.7264
Epoch 67/100
48/48 ————— 0s 5ms/step - loss: 5.1121 - mean_absolute_error: 5.1121 - val_loss: 4.5076 - val_mean_absolute_error: 4.5076
Epoch 68/100
48/48 ————— 0s 5ms/step - loss: 5.1145 - mean_absolute_error: 5.1145 - val_loss: 4.3360 - val_mean_absolute_error: 4.3360
Epoch 69/100
48/48 ————— 0s 5ms/step - loss: 4.5257 - mean_absolute_error: 4.5257 - val_loss: 4.2487 - val_mean_absolute_error: 4.2487
Epoch 70/100
48/48 ————— 0s 5ms/step - loss: 5.5188 - mean_absolute_error: 5.5188 - val_loss: 4.4513 - val_mean_absolute_error: 4.4513
Epoch 71/100
48/48 ————— 0s 5ms/step - loss: 5.2237 - mean_absolute_error: 5.2237 - val_loss: 5.7062 - val_mean_absolute_error: 5.7062
Epoch 72/100
48/48 ————— 0s 6ms/step - loss: 4.9939 - mean_absolute_error: 4.9939 - val_loss: 4.9649 - val_mean_absolute_error: 4.9649

Epoch 73/100
48/48  0s 5ms/step - loss: 5.8261 - mean_absolute_error: 5.8261 - val_loss: 6.5515 - val_mean_absolute_error: 6.5515
Epoch 74/100
48/48  0s 5ms/step - loss: 5.5228 - mean_absolute_error: 5.5228 - val_loss: 4.9510 - val_mean_absolute_error: 4.9510
Epoch 75/100
48/48  0s 5ms/step - loss: 4.7113 - mean_absolute_error: 4.7113 - val_loss: 4.8822 - val_mean_absolute_error: 4.8822
Epoch 76/100
48/48  0s 6ms/step - loss: 4.9737 - mean_absolute_error: 4.9737 - val_loss: 4.5091 - val_mean_absolute_error: 4.5091
Epoch 77/100
48/48  0s 5ms/step - loss: 4.8898 - mean_absolute_error: 4.8898 - val_loss: 4.9352 - val_mean_absolute_error: 4.9352
Epoch 78/100
48/48  0s 5ms/step - loss: 5.1666 - mean_absolute_error: 5.1666 - val_loss: 4.1798 - val_mean_absolute_error: 4.1798
Epoch 79/100
48/48  0s 8ms/step - loss: 5.1114 - mean_absolute_error: 5.1114 - val_loss: 4.7328 - val_mean_absolute_error: 4.7328
Epoch 80/100
48/48  1s 8ms/step - loss: 4.7451 - mean_absolute_error: 4.7451 - val_loss: 4.5289 - val_mean_absolute_error: 4.5289
Epoch 81/100
48/48  0s 10ms/step - loss: 5.3272 - mean_absolute_error: 5.3272 - val_loss: 5.8869 - val_mean_absolute_error: 5.8869
Epoch 82/100
48/48  1s 10ms/step - loss: 5.8421 - mean_absolute_error: 5.8421 - val_loss: 4.9204 - val_mean_absolute_error: 4.9204
Epoch 83/100
48/48  0s 9ms/step - loss: 5.0729 - mean_absolute_error: 5.0729 - val_loss: 5.1179 - val_mean_absolute_error: 5.1179
Epoch 84/100
48/48  0s 5ms/step - loss: 5.1582 - mean_absolute_error: 5.1582 - val_loss: 4.7741 - val_mean_absolute_error: 4.7741
Epoch 85/100
48/48  0s 5ms/step - loss: 5.9327 - mean_absolute_error: 5.9327 - val_loss: 5.0899 - val_mean_absolute_error: 5.0899
Epoch 86/100
48/48  0s 6ms/step - loss: 4.5545 - mean_absolute_error: 4.5545 - val_loss: 4.5173 - val_mean_absolute_error: 4.5173
Epoch 87/100
48/48  0s 5ms/step - loss: 4.6914 - mean_absolute_error: 4.6914 - val_loss: 5.2715 - val_mean_absolute_error: 5.2715
Epoch 88/100
48/48  0s 5ms/step - loss: 5.5674 - mean_absolute_error: 5.5674 - val_loss: 5.0540 - val_mean_absolute_error: 5.0540
Epoch 89/100
48/48  0s 5ms/step - loss: 5.3899 - mean_absolute_error: 5.3899 - val_loss: 5.3250 - val_mean_absolute_error: 5.3250
Epoch 90/100
48/48  0s 6ms/step - loss: 5.4996 - mean_absolute_error: 5.4996 - val_loss: 4.3373 - val_mean_absolute_error: 4.3373

```

Epoch 91/100
48/48 ————— 0s 5ms/step - loss: 4.5817 - mean_absolute_error:
4.5817 - val_loss: 4.4781 - val_mean_absolute_error: 4.4781
Epoch 92/100
48/48 ————— 0s 5ms/step - loss: 4.5650 - mean_absolute_error:
4.5650 - val_loss: 4.3282 - val_mean_absolute_error: 4.3282
Epoch 93/100
48/48 ————— 0s 6ms/step - loss: 4.4330 - mean_absolute_error:
4.4330 - val_loss: 5.0588 - val_mean_absolute_error: 5.0588
Epoch 94/100
48/48 ————— 0s 5ms/step - loss: 5.0841 - mean_absolute_error:
5.0841 - val_loss: 5.2926 - val_mean_absolute_error: 5.2926
Epoch 95/100
48/48 ————— 0s 5ms/step - loss: 5.9248 - mean_absolute_error:
5.9248 - val_loss: 5.0346 - val_mean_absolute_error: 5.0346
Epoch 96/100
48/48 ————— 0s 5ms/step - loss: 4.8770 - mean_absolute_error:
4.8770 - val_loss: 4.7090 - val_mean_absolute_error: 4.7090
Epoch 97/100
48/48 ————— 0s 6ms/step - loss: 5.1239 - mean_absolute_error:
5.1239 - val_loss: 4.4569 - val_mean_absolute_error: 4.4569
Epoch 98/100
48/48 ————— 0s 5ms/step - loss: 4.5058 - mean_absolute_error:
4.5058 - val_loss: 5.3546 - val_mean_absolute_error: 5.3546
Epoch 99/100
48/48 ————— 0s 6ms/step - loss: 4.6462 - mean_absolute_error:
4.6462 - val_loss: 5.0824 - val_mean_absolute_error: 5.0824
Epoch 100/100
48/48 ————— 0s 5ms/step - loss: 4.7164 - mean_absolute_error:
4.7164 - val_loss: 4.8538 - val_mean_absolute_error: 4.8538

```

```

In [75]: # STEP 11 : PREDICTING
         prediction = NN_model.predict(X_test)

```

```

10/10 ————— 0s 15ms/step

```

```

In [77]: # STEP 12: DISTRIBUTION OF ERRORS
         plt.figure(figsize=(12,6))
         sns.distplot((y_test.values.reshape(-1,1) - prediction))
         plt.title("Error Distribution")
         plt.show()

```

/tmp/ipython-input-1539574209.py:3: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

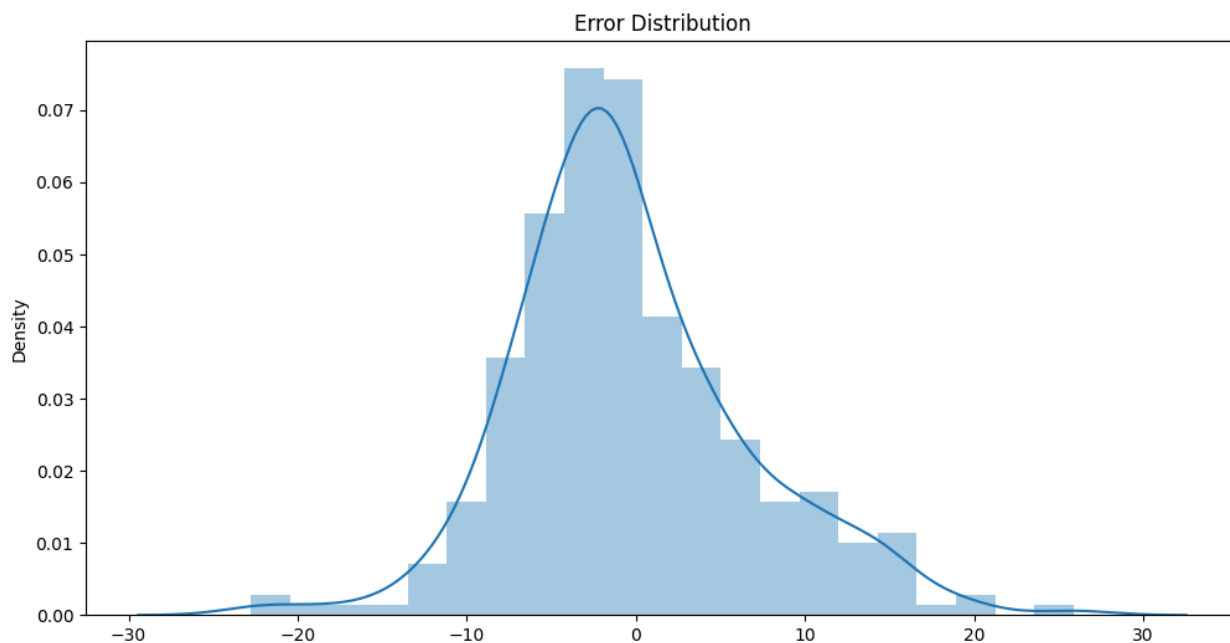
Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

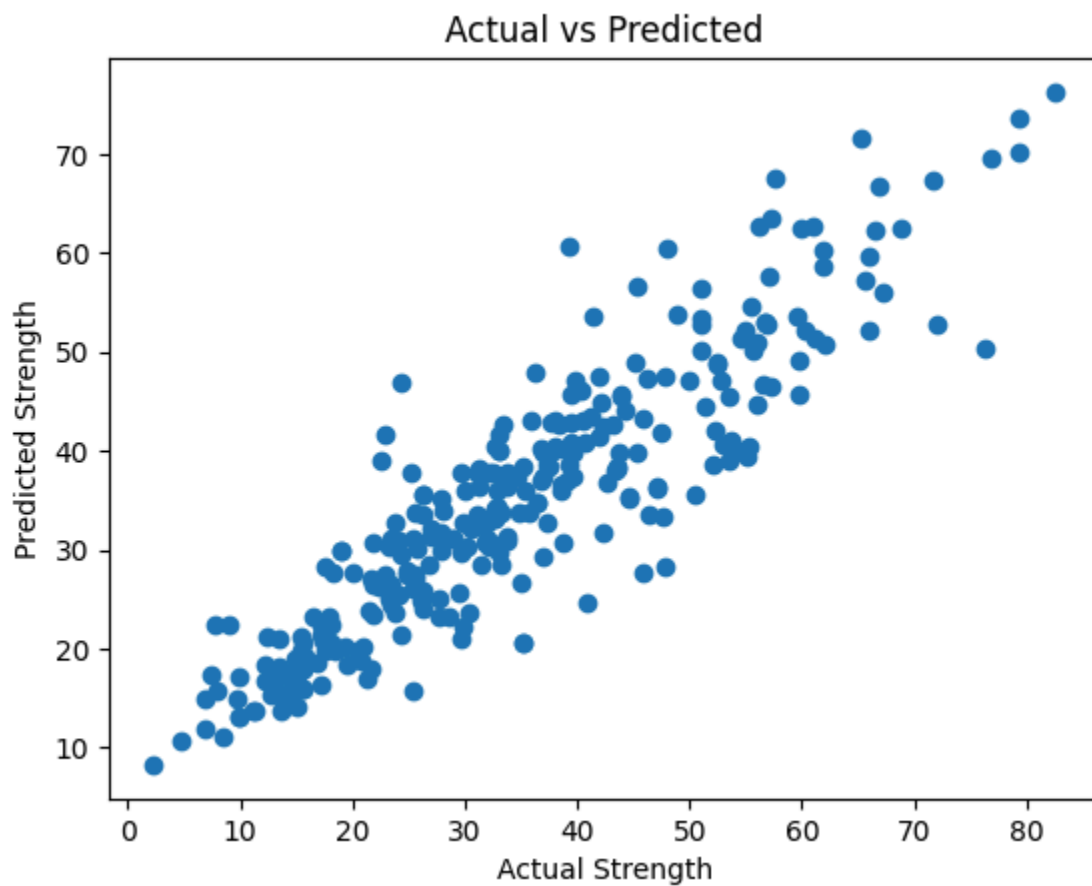
```

sns.distplot((y_test.values.reshape(-1,1) - prediction))

```



```
In [80]: # STEP 13: SCATTERPLOT OF ACTUAL VS PREDICTED
plt.scatter(y_test, prediction)
plt.xlabel("Actual Strength")
plt.ylabel("Predicted Strength")
plt.title("Actual vs Predicted")
plt.show()
```



```
In [81]: # STEP 14: EVALUATION METRICS
MAE = mean_absolute_error(y_test, prediction)
MSE = mean_squared_error(y_test, prediction)
RMSE = np.sqrt(MSE)

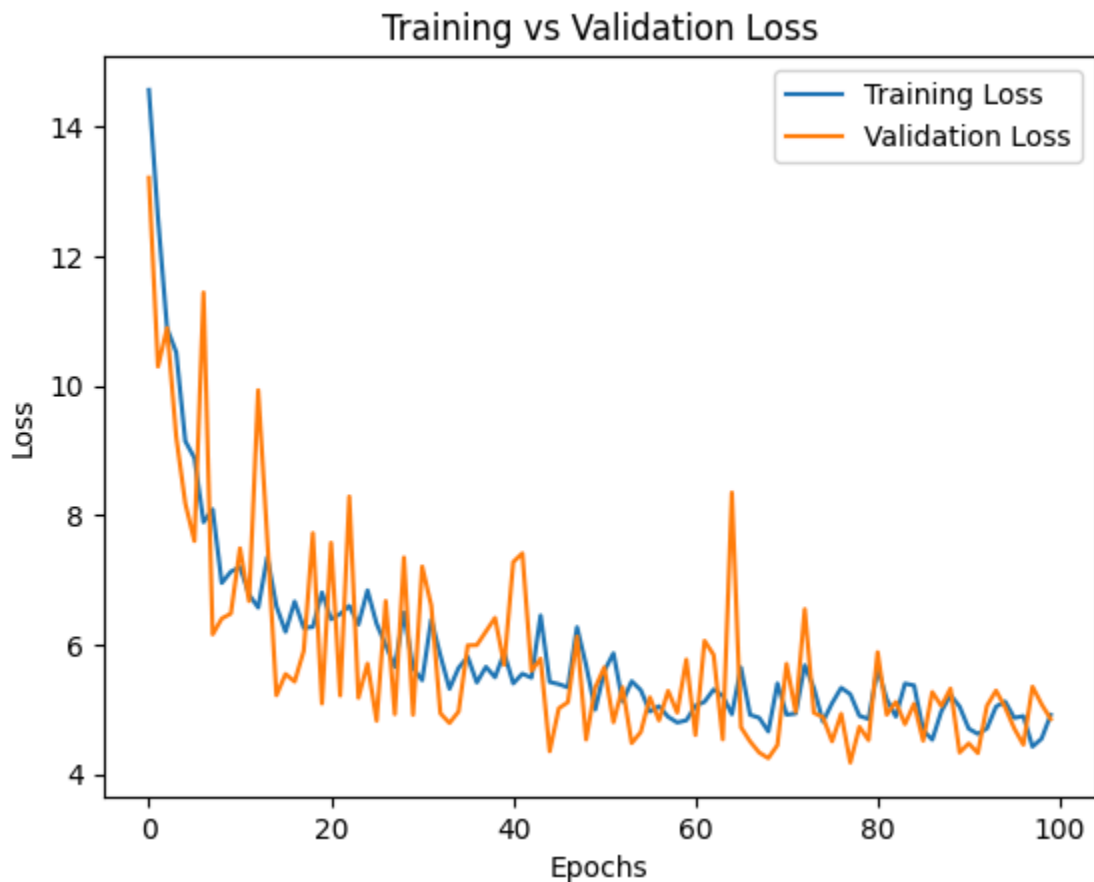
print("MAE:", MAE) # average absolute error
print("MSE:", MSE) # squared error
print("RMSE:", RMSE) # root mean squared error
```

```
MAE: 5.318952966652169
MSE: 48.57370119260867
RMSE: 6.9694835671381465
```

```
In [82]: # STEP 20: TRAINING vs VALIDATION LOSS GRAPH
train_loss = model_history.history['loss'] # training loss
val_loss = model_history.history['val_loss'] # validation loss

plt.plot(train_loss)
plt.plot(val_loss)

plt.xlabel("Epochs")
plt.ylabel("Loss")
plt.title("Training vs Validation Loss")
plt.legend(["Training Loss", "Validation Loss"])
plt.show()
```



-Final Conclusion

- ANN model predicted concrete strength successfully.
- Training and validation loss reduced properly.
- Predicted values are close to actual values.
- Low MAE and RMSE show good model accuracy.

In []: